

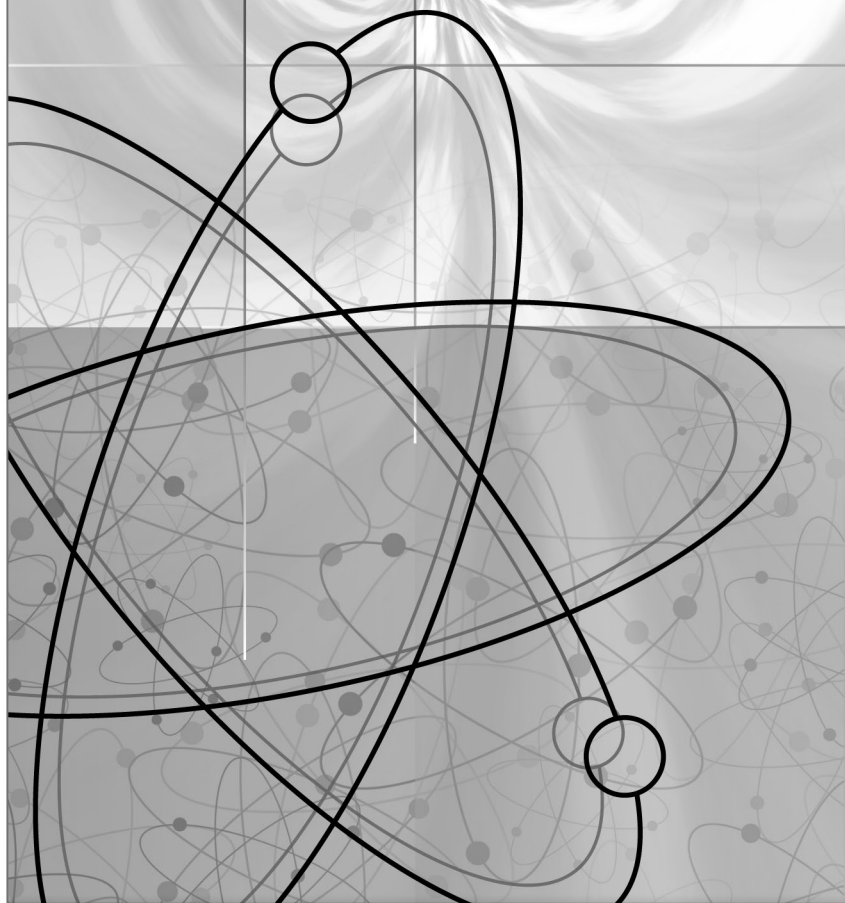
Interpreting data in Senior Physics

Grahame Dobinson



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Interpreting data in **Senior Physics**



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Grahame Dobinson is a science teacher with many years of experience in teaching all branches of science. He has a genuine love of this subject area, and is an established and successful author who has a longstanding association with Blake Education.

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INTRODUCTION

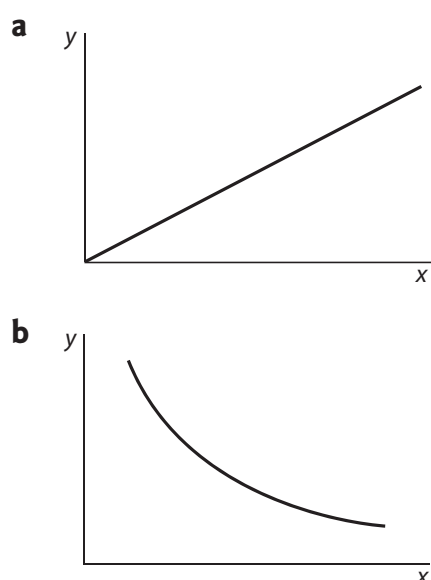
A common method of representing data from an experimental investigation is to draw a graph of the data in order to see if any pattern can be determined. A definite pattern can then be expressed as a mathematical relationship, allowing further predictions.

Aim

To observe and graph some physical data from experiments so that patterns can be identified and hence relationships can be found.

Data: Set 1

Two commonly occurring graphical shapes resulting from the plotting of physical data are shown below.



The relationships indicated by these graphs between the independent variable (x) and the dependent variable (y) are expressed in words respectively as (a) y is directly proportional to x and (b) y is inversely proportional to x .

Analysing the data

1 What is the mathematical way of expressing each of these relationships using symbols rather than words?

a _____

b _____

Note: The data for sets 2 and 3 relate to data collected from two separate experiments.

Data: Set 2

In an experiment to see if there is a relationship between the volume of a mass of gas and the pressure acting on it, a student used a syringe with a plunger to see if the volume of the trapped gas changed as the pressure on it

was increased by pushing in the plunger. The apparatus used is shown in Figure 1 and the results of six readings are listed in Table 1.

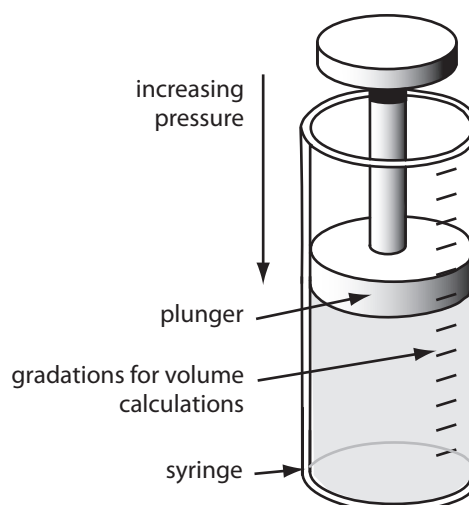


Figure 1

Table 1

Pressure (units)	Volume (units)	$\frac{1}{V}$	PV
2.0	20.0		
4.0	10.0		
6.0	6.7		
8.0	5.0		
10.0	4.0		
12.0	3.3		

Analysing the data

2 Draw a graph of volume vs pressure.

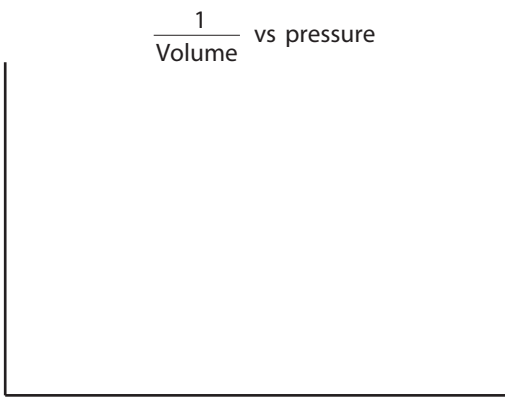
Volume vs pressure



3 Is this graph a straight line? _____

4 Calculate the values of $\frac{1}{V}$ and enter these values in the third column of the table.

5 Draw a second graph of $\frac{1}{V}$ vs P .



Interpreting the data

- 6 Does the graph in question 4 appear to be a straight line?
- 7 Give a mathematical relationship that exists between P and V .
- 8 What does this relationship mean about the product PV for a gas?
- 9 Check this relationship by filling in the last column in the table. Are the values consistent with what has been said in question 7?

Data: Set 3

An experiment is carried out to see if a relationship exists between the intensity of light and the distance from the light source. The apparatus used is shown in Figure 2 and the results are listed in Table 2.

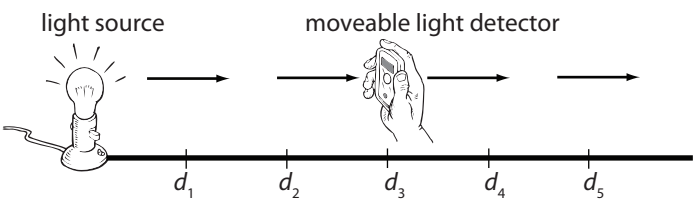


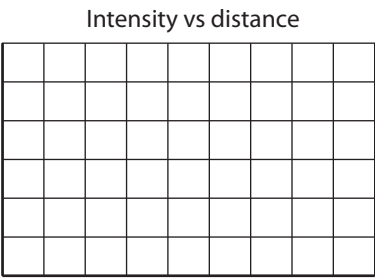
Figure 2

Table 2

Distance (m)	Intensity (units)		
1.0	500		
2.0	125		
3.0	56		
4.0	31		
5.0	20		

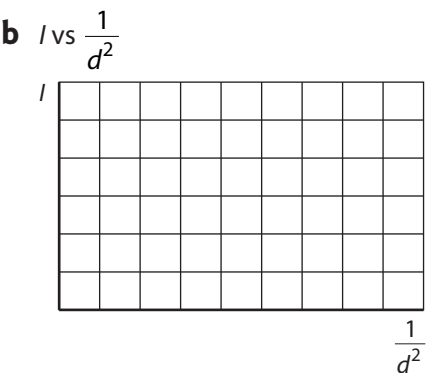
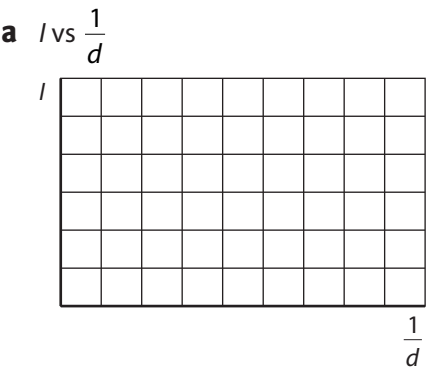
Analysing the data

10 Draw a graph of intensity vs distance.



Is this graph a straight line? _____

11 In the third and fourth columns of the table, insert calculated values for $\frac{1}{\text{distance}}$ and $\frac{1}{(\text{distance})^2}$, respectively, and then draw two more graphs of:



Interpreting the data

- 12 Is either of these graphs a straight line? Identify the graph(s).
- 13 Suggest a relationship that exists between intensity and distance.

2 Uncertainty in measurement

INTRODUCTION

Whenever a measurement is made, a 'perfectly accurate' result is never achieved. Instead, the measurement is limited by the smallest subdivision (or least count) of the measuring instrument.

Aim

To see how the combining of measurements is affected when these limitations in measurement are themselves combined, specifically in addition and subtraction.

Data: Set 1 A single measurement

A length of pipe is measured against a ruler that is marked in 1 mm units as its smallest division (or least count). The measurement of that length is shown in Figure 1.

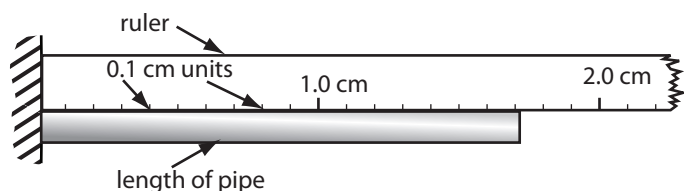


Figure 1

Analysing the data

The measurement of the length of pipe is recorded as 1.7 cm, since the end of the pipe is closest to the 1.7 mark on the ruler.

- 1 If the pipe had aligned at either position B or C, as shown in Figure 2, would the measurement still be 1.7?

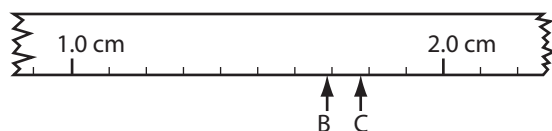


Figure 2

B _____ C _____

- 2 What are the lower and the upper limits for a reading of 1.7 cm to be obtained? Note: Here the least count can be subdivided with the eye.

a Lower _____

b Upper _____

- 3 The range of readings covered by 1.7 cm can be written as $1.7 \pm$ _____

This ' $\pm$ ' value is called the 'uncertainty' of the single measurement, as it covers the range of possibilities of measurements that may be correct.

Data: Set 2 Adding measurements

Suppose that two pieces of pipe, each measuring 1.7 ± 0.05 cm are added end to end as shown in Figure 3.

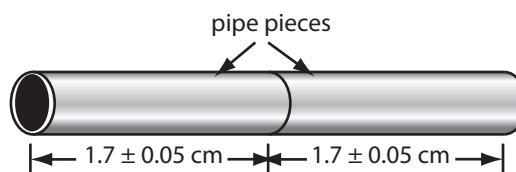


Figure 3

It is obvious that the combined length is 3.4 cm, but what is the uncertainty? This is answered as follows:

- 4 What is the minimum possible combined length of the pipes? _____
- 5 What is the maximum possible combined length of the pipes? _____
- 6 Now write the combined length and include the new uncertainty so that the minimum and maximum possibilities are included.

Data: Set 3 Subtracting measurements

Another piece of pipe is measured as being 3.4 ± 0.05 cm long. It is cut cleanly into two (without loss) and one new piece is measured to be 1.9 ± 0.05 cm. Obviously the second piece is 1.5 cm long.

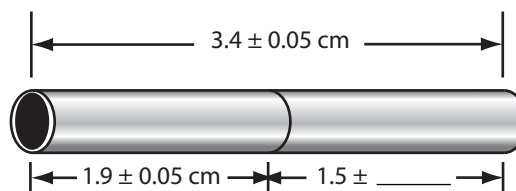


Figure 4

- 7 What is the uncertainty? First, find the minimum possible length of the second piece, considering the uncertainties. _____
- 8 What is the maximum possible length of the second piece? _____
- 9 Now write the length of the 'second piece' including the new uncertainty. _____

Interpreting the data

- 10 Some simple rules emerge from these examples and they can be applied to all similar situations. They can be stated by completing the follow sentences:

- a The uncertainty for a single measurement is _____.
- b When two measurements are added, their uncertainties _____ to give the uncertainty of their sum.
- c When one measurement is subtracted from another, their uncertainties _____ to give the uncertainty of their remainder.

INTRODUCTION

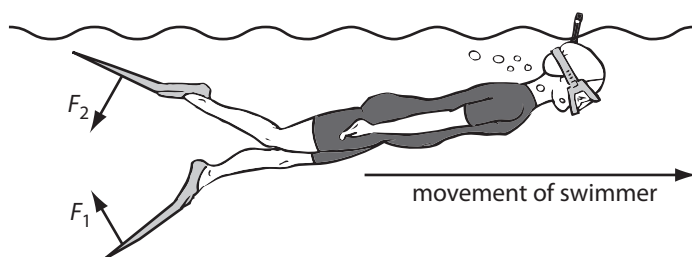
When a person dons fins (flippers), mask, snorkel, etc. and sets out to enjoy the underwater marvels, there is quite a bit of Physics involved in their enjoyment.

Aim

To consider some of the Physics issues, particularly forces, involved when a person swims underwater.

Data

A woman with all the necessary equipment and of total mass 70 kg swims horizontally at a shallow depth of 1.0 m while holding her breath. The force that propels her is supplied solely by the movement of her fins. She moves leisurely through the water with a velocity of 0.6 m s^{-1} . At a particular instant her legs are positioned such that the fins are both angled at 30° to the horizontal, one moving downwards and one moving upwards as she does a modest 'scissors' kick. The forces F_1 and F_2 are both equal to 50.0 N. The value of a_g can be taken as 10 m s^{-2} for these calculations. The diagram below shows the situation.



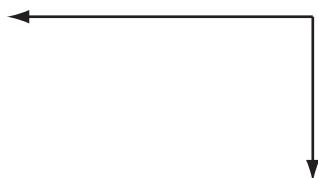
Analysing the data

1 Resolve the forces F_1 and F_2 into a vertical and a horizontal component graphically.

a Resolving F_1



b Resolving F_2



2 Using the convention that to the right horizontally and towards the top of the page vertically are the '+' directions, what is the vector sum of the vertical and the horizontal components of F_1 and F_2 respectively?

a F_1 _____

b F_2 _____

3 The horizontal resultant forces of both F_1 and F_2 are to the left and yet the swimmer moves to the right. Explain how this comes about.

4 The force making the swimmer move to the right does not make her accelerate. Instead, she moves with a constant velocity of 0.6 m s^{-1} . How can this be explained?

5 The swimmer plus her equipment has a total mass of 70 kg. What force, in newtons, does gravity exert on this mass?

Some additional data

It was the famous mathematician and scientist Archimedes who proposed that objects float because there is an 'upthrust' force equal to the *weight* of water displaced by the object. When this 'upthrust' force equals the downward weight force, the object doesn't move up or down – it floats.

6 If the swimmer experiences no upwards or downwards net force at 1 metre depth, what must be the actual size of the upwards force of the water on the swimmer?

Interpreting the data

7 If the swimmer puts on scuba gear that has a mass of 5 kg, what additional downward force is added to her weight force?

8 What additional upwards force is needed to counter the force in question 7?

9 The scuba gear would, because of its volume, displace extra water. What effect would this have on the upthrust force, according to Archimedes?

10 When a swimmer is in salt water, the upthrust force is greater than when the swimmer is in fresh water (which is less dense). Suggest a reason why the force is greater.

INTRODUCTION

The motion of an object simply sliding down an inclined plane can reveal relationships that exist between the parameters of the motion.

Aim

To find out if any relationship exists between the time of travel of an object as it slides down a 'frictionless' inclined plane and the distance covered.

Data

An experiment was carried out with a long plane and an electronic timing device. Distances were marked along the plane and the total distance moved by the object was recorded at half-second intervals. The experiment was repeated a number of times and the average distance travelled for each time interval was recorded. The apparatus used is shown in Figure 1. A table of results is also given.

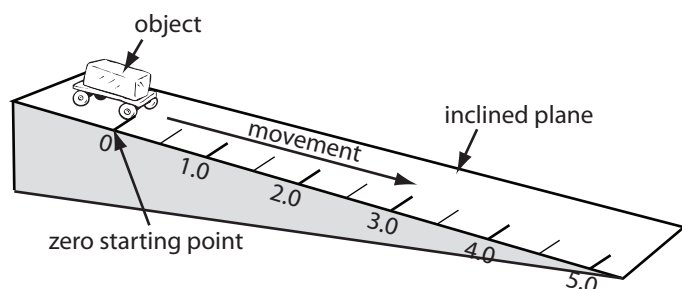


Figure 1

Total time of travel (s)	0.50	1.00	1.50	2.00	2.50	3.00
Average distance (m)	0.125	0.500	1.125	2.000	3.125	4.500
(Time) ²						

Analysing the data

- 1** The independent variable here is 'time', which is set at 0.50 s intervals. Construct a graph that plots the two variables on the correct axes.



Interpreting the data

- 2** Does the graph show a direct relationship between the two variables?

- 3** Calculate the value of (time)² for each of the six recordings and enter the values in the table.
- 4** Draw a second graph of distance vs (time)² and draw a 'line of best fit'.



- 5** Is this graph a straight line? _____
- 6** Write an expression that relates distance to time in this experiment.

- 7** What is the value of the gradient in the second graph?

- 8** Given that one equation of motion is $s = ut + \frac{1}{2}at^2$, what does the gradient of the second graph represent?

- 9** From the above determine the acceleration of the object down the plane.

- 10** From one of the graphs predict **a** how far the object would have moved in 2.30 s and **b** how long it would take the object to move 4.0 m down the plane.
a _____
b _____
- 11** Since the object is made to slide in a set, unchanging direction, what term could be used to better describe the 'distance' the object moves?

- 12** From the information revealed by the graphs, could the speed of the object at the end of 3.00 s be calculated? _____
Explain. _____
- 13** If the speed of the object at the end of 3.00 s was known, what additional information would be needed before its kinetic energy could be determined?

- 14** Suggest what would happen to the distance travelled in the same time intervals if the angle of the plane to the horizontal were to be increased.

INTRODUCTION

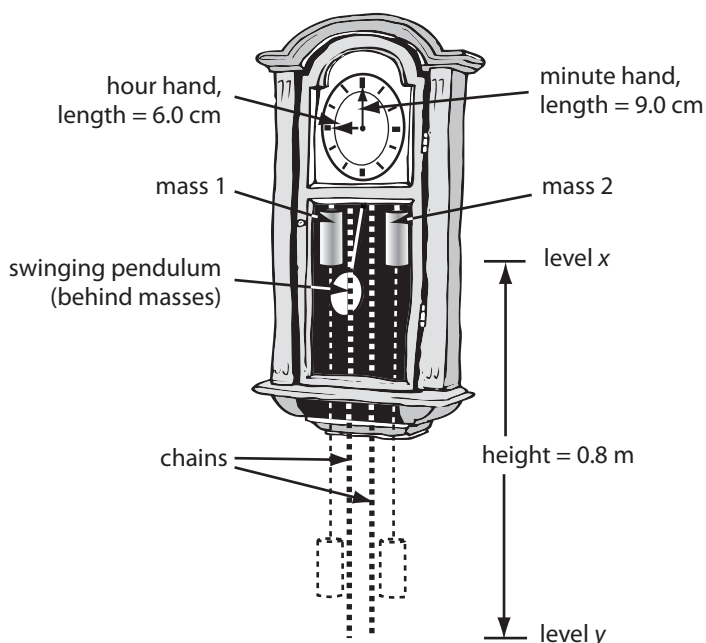
Most modern clocks are powered by electricity, either by the household power supply or by batteries. However, older clocks were usually powered by some form of potential energy.

Aim

To investigate and interpret some motion and energy data about a chiming clock.

Data

The large wall clock shown works using gravitational potential energy (PE). The two masses, each of 2.0 kg, are pulled up to level X using the light chains. As they move slowly down to level Y under their own weight, they cause the pendulum to swing regularly from side to side and so drive the clock mechanism. The clock chimes every half-hour. It takes exactly seven days for the two masses to reach level Y, at which stage the clock stops until the masses are raised to level X again. Use $g = 9.8 \text{ m s}^{-2}$.



Interpreting the data

1 Determine the gain in PE when the two masses are raised from level Y to level X.

2 Assume that 80% of the PE is used to drive the hands of the clock as the masses descend. How much energy does this represent for a seven-day period?

3 Account for the other 20% of energy not used to drive the clock hands.

4 Calculate the overall rate of loss of gravitational PE in the seven-day period in joules per second.

5 What is another physical term for the 'rate of energy consumption' by the clock? (Give the term and its units.)

6 If the length of the minute hand is as shown in the diagram, through what distance does the tip of the hand move during one revolution?

7 Determine the speed of the minute hand as the clock operates.

8 Would it be true to say that the 'speed' and the 'velocity' of the tip of the minute hand are the same as the clock operates?

Explain.

9 Calculate the total distance moved by the tip of the minute hand over a seven-day period.

10 Show on the axes supplied the shape of a distance vs time graph that represents an hour's movement of the tip of the minute hand.



11 What is the displacement of the tip of the minute hand after one hour?

12 After what period(s) of time is the displacement of the tip of the minute hand at a maximum?

INTRODUCTION

The force of friction is one that always opposes motion. It may be large or small depending on a number of factors, especially the nature of the surfaces in contact. A coefficient of friction is a number to indicate the 'stickiness' of two surfaces as one is made to slide across the other. A small coefficient indicates that the surfaces slide easily; a large coefficient indicates that the frictional force between the two prevents one from sliding easily over the other.

Aim

To take suitable measurements that will allow the coefficient of friction to be determined for two pieces of similar wood.

Data

An experiment is set up as shown (Figure 1) to measure the force required to overcome the frictional force opposing the motion of the block. The mass of the block is 1.0 kg. Masses are placed on the block and the pulling force to overcome friction is measured. The pulling force is measured just as the block starts to move, so the pulling force is equal in size to the frictional force. The results are shown in the table.

Note: The value of g can be taken as 10 m s^{-2} .

Experiment

Mass being pulled (kg)	Weight of each mass (N)	Reaction force (F_r) (N)	Frictional force F_f (N)	$F_f:F_r$
1.0			5.0	
1.5			7.6	
2.0			10.1	
2.5			12.5	
3.0			14.9	
3.5			17.6	

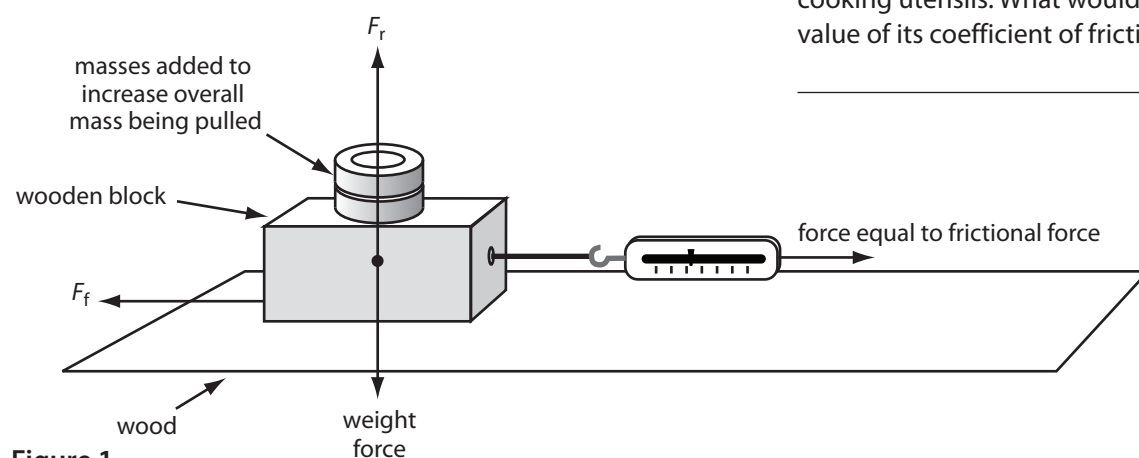


Figure 1

Analysing the data

- 1 Calculate the values where necessary and complete the table.
- 2 What can be said about the values for the last column of the table?

- 3 This value is called the 'coefficient of starting friction' for the two surfaces and is given the symbol μ . The value decreases a little once the object is moving. Write a formula linking μ to the forces from which it can be calculated.

- 4 Using the coefficient of (starting) friction calculated here, predict:
 - a the frictional force when the mass being pulled is 5.0 kg.

 - b the mass being moved when a frictional force of 16.0 N is present.

Interpreting the data

- 5 For two polished metal surfaces, μ has a value of 0.13. What would be the frictional force between the surfaces if a 2.5 kg polished metal block just starts to move over another polished metal surface?

- 6 The fact that the coefficient of (starting) friction is greater than the coefficient of (sliding) friction explains why 'antilock' brakes are more effective than those that allow wheels to lock and skid. Explain.

- 7 'Teflon' is the anti-stick coating used on many cooking utensils. What would this imply about the value of its coefficient of friction?

INTRODUCTION

Many experiments to determine the acceleration due to gravity have used fairly complicated pieces of equipment and expensive timing devices. The following one is simple and uses basic items of equipment.

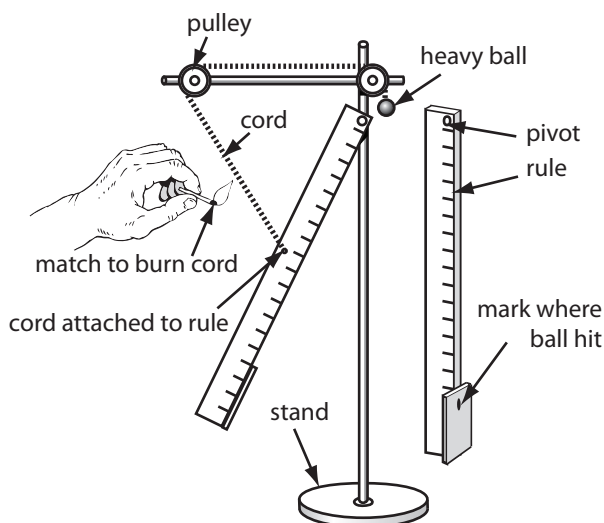


Figure 1

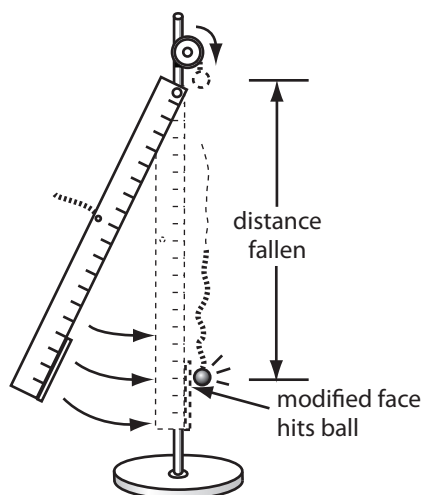


Figure 2

Aim

To obtain a value for the acceleration (a_g) due to gravity using the simple benchtop apparatus shown in Figure 1.

Data

The apparatus uses a sturdy stand, some light cord, a modified metre rule, a couple of pulleys, a heavy ball attached to the cord and a stopwatch. The rule is modified slightly as shown. The cord suspending the ball is passed over the pulleys and connected to the metre rule. When the cord is burned through, the ball starts to fall and the rule starts to swing. As the rule reaches the vertical, it collides with the ball, which leaves a mark on the modified face of the rule, indicating how far it has fallen. Since initial velocity and distance fallen by the ball are now known, it only requires the time of fall of the ball to be determined and the acceleration due to gravity can be

calculated. This time can be determined from the time of swing of the rule, since the two times are the same. After repeat trials have been carried out and averages calculated, a student records the following results:

- Time of fall of ball = _____ s
- Distance fallen by ball = 0.75 m
- Initial velocity of ball = 0.0 m s^{-1}

Analysing the data

- 1 What fraction of the periodic time (T) of the metre rule's swing is the same as the time of fall of the ball?

- 2 The student records the time for 10 complete oscillations (swings) of the rule as 15.6 s. From this, determine the time of fall of the ball and enter this in the space above.

- 3 With this additional piece of information and a selected equation of motion, calculate the value of a_g .

Interpreting the data

- 4 How does the calculated value for a_g in question 3 compare with the accepted value of 9.8 m s^{-2} ?

- 5 Considering the apparatus used, suggest a couple of reasons why the exact value is not achieved.
a _____
b _____
- 6 The recognised value for a_g on the Moon's surface is approximately 1.6 m s^{-2} . If the experiment were to be carried out on the Moon's surface, would the values described below increase or decrease?
a the time it takes for the ball to fall a distance of 0.75 m

- b the period of swing of the pendulum

- 7 The Moon has no atmosphere (or air). How would this affect the time of swing of the pendulum compared to when an atmosphere is present?

- 8 If a ball were allowed to fall 0.75 m on the Moon, how long would it take to do this?

INTRODUCTION

There have been many methods devised throughout history for determining a_g , the acceleration due to gravity.

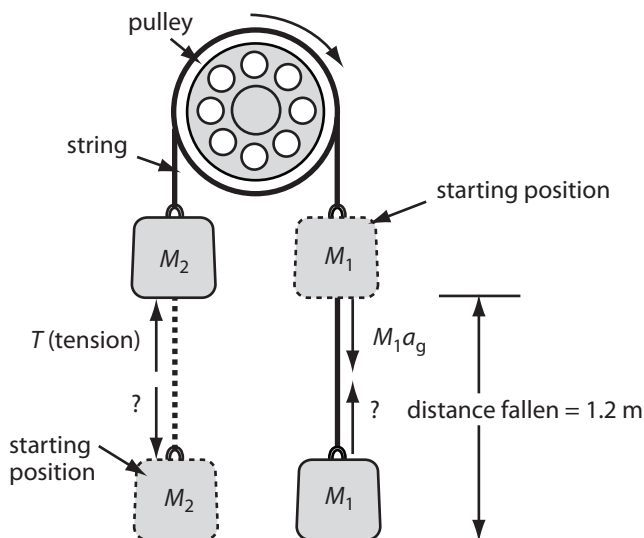
Aim

To use a particular method, first used a long time ago, to determine a value for a_g and to compare the value obtained with the modern accepted value.

Data

The apparatus used consists of two masses connected by a string over a pulley as shown. The mass M_1 is slightly greater than the mass M_2 . As a result, the larger mass, when allowed to move, moves down while the smaller mass moves up. The distance moved down by M_1 is measured and the time taken for its movement is recorded. This allows the acceleration of M_1 to be calculated and hence the value of g to be determined. The apparatus used is shown in the diagram and the average time for M_1 to descend the 1.2 m distance is given.

- Distance moved by $M_1 = 1.2$ m
- Average time taken = 3.2 s
- Mass sizes: $M_1 = 420$ g, $M_2 = 400$ g



Analysing the data

- Label the two unidentified forces acting on the system as shown.
- Write an equation of motion for the downwards movement of M_1 (i.e. in the form of $F_n = ma$).

Rewrite this equation making T (tension in the string) the subject of the formula.

- Write an equation of motion for the upwards movement of M_2 .

Rewrite this equation, again making T the subject of the formula.

- Since T has a constant value, the rewritten equations from question 2 and question 3 can be equated and so T is eliminated. Do this and so devise an equation with g as the subject of the formula in terms of M_1 , M_2 and a (the acceleration of the masses).

- Mass M_1 , starting from rest, has moved a distance of 1.2 m in a time of 3.2 s. Calculate its acceleration, a , over the 1.2 m distance.

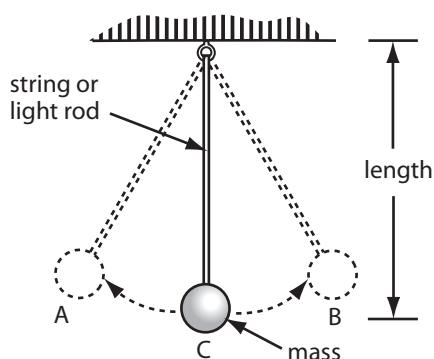
- Now that a has been determined, its value can be used in the equation devised in question 4. Do this and so determine g , the acceleration due to gravity.

Interpreting the data

- The accepted value for g is 9.8 m s^{-2} .
 - Is the calculated value smaller than or larger than the accepted value?
 - Suggest a reason why this might be so, considering the apparatus used.
- The experiment is repeated using $M_2 = 400$ g, but M_1 is increased to 440 g. How would this affect each of the following values (i.e. increase it, decrease it or leave it unchanged)?
 - the time of fall of M_1
 - the acceleration of M_1
 - the value of g

INTRODUCTION

A simple pendulum may consist of a piece of string or light rod supporting a mass that is allowed to swing from side to side. A child's swing is, in effect, a simple pendulum. The pendulum has been used as a timing device in clocks for a very long time.



Aim

To determine whether or not the *length* of a simple pendulum affects the *period* of swing of the pendulum.

Data

The period (T) of the pendulum is the time taken for one complete 'oscillation', with the pendulum ending up where it started. In an experiment, a student used the same mass and made the pendulum move from side to side the same distance for every measurement she made. She gradually increased the length of the pendulum and timed 10 complete swings for each length. She then found the average time for each length. Her results are shown in the table below.

Length of pendulum (m)	Average period of swing (s)	(Period) ²
0.4	1.27	
0.6	1.56	
0.8	1.80	
1.0	2.01	
1.2	2.20	
1.4	2.38	
1.6	2.54	
1.8	2.70	
2.0	2.84	
2.2	2.98	

Analysing the data

- 1 Plot a graph of period vs length on the axes provided.

Period vs length

- 2 a Is the graph in question 1 a straight line?

- b Can it be said that period $\propto$ length?

- 3 Calculate values of (period)² and enter them in the table.

- 4 Plot a graph of (period)² vs length and decide whether or not the relationship can be represented by a straight line. Yes or No?

(Period)² vs length

Interpreting the data

- 5 Compare the two graphs and write down a mathematical relationship that links 'period' and 'length'.

- 6 Rewrite the relationship for question 5 so that 'period' is the subject.

- 7 What force is responsible for the continued motion of the pendulum?

- 8 As the pendulum swings from side to side there are always two forces acting on it to cause its motion. What are these two forces?

- 9 As the pendulum passes through position C (i.e. when it is vertical), what is the *net* force acting on it?

- 10 At what position(s) would the pendulum have its greatest:

a potential energy? _____

b velocity? _____

INTRODUCTION

Handicap systems are used in many sports and take many different forms. This exercise looks at two different systems commonly used for races such as those in athletics and cycling. In foot races such as the professional sprints the Botany Gift and the Stawell Gift, the slower sprinters are given slightly shorter distances to run, depending on their individual skills. In cycling, the slower contestants are given earlier starts, i.e. time advantages, and all contestants race over the same distance. The faster riders have to catch up to those who started earlier.

Aim

To look at some data from two such events and to use the data to determine aspects such as distance covered, time taken and average velocity for the participants in those events.

Data: Set 1

Four sprinters are considered in this race over 100 m. Their starts are shown in Figure 1.

Lane 2 wins in a time of 9.80 s. Lane 3, the fastest or 'scratch' sprinter, is second and is 0.05 s behind the winner. Lanes 1 and 4 'dead heat' for third, another 0.05 s behind second place.

Interpreting the data

Note: The answers to the questions can be expressed to two decimal places.

- 1 Calculate the average velocity of:
 - a the winner. _____
 - b the second placegetter. _____
- 2 When the winner reaches the finish line, how far behind is the scratch runner? _____
- 3 Determine how much further the first two runners would have to run for the scratch runner to catch up to the winner. (Assume average velocities are maintained.) _____

1.5 m	lane 1	equal 3rd
3.0 m	lane 2	winner in 9.80 s
scratch	lane 3	second
4.0 m	lane 4	equal 3rd

Figure 1

- 4 Calculate the average velocities of the other two runners.
 - a Lane 1: _____
 - b Lane 4: _____
- 5 Rank the four runners in this race on their average velocities from fastest to slowest. Use lane numbers. _____

Data: Set 2

Four competitors in a 20 km cycling road race are considered here. The slowest rider R_1 has a 60 s start over the scratch rider R_4 . The other two riders R_2 and R_3 have starts of 40 s and 20 s respectively. The starter starts to count back from 65 and each rider starts when his/her time is called. When the starter works down to '... 3, 2, 1, 0', the scratch rider starts.

The scratch rider R_4 averages a velocity of 50 km h^{-1} and wins the race by 5.0 s from R_3 . Third place, R_1 , is a further 8.0 s away; R_2 is unplaced.

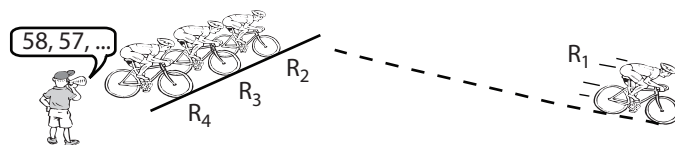


Figure 2

Analysing the data

- 6 What is the average speed of the scratch rider in m s^{-1} ? _____
- 7 Calculate the time the scratch rider took for the race in minutes. _____
- 8 What time, in seconds, did riders R_3 and R_1 take for the race?

R_3 _____

R_1 _____
- 9 Calculate the average velocity (in m s^{-1}) of R_3 for the race. _____

INTRODUCTION

The Law of Conservation of Energy is one which accounts for energy as it changes from one form to another. It means, basically, that an original amount of energy can be accounted for no matter what other form(s) it may be changed into. That is, energy does not simply 'disappear' from any situation. Such accounting can be seen in the following examples.

Aim

To investigate two situations in which work and energy are involved in relation to the Law of Conservation of Energy.

Data: Work and kinetic energy

Consider a situation in which a stationary mass on a 'frictionless' surface has a force applied to it. The force (F) acting on the mass (m) over a distance (s) produces an acceleration (a) that makes the mass achieve a final velocity (v) at the end of the distance (Figure 1).

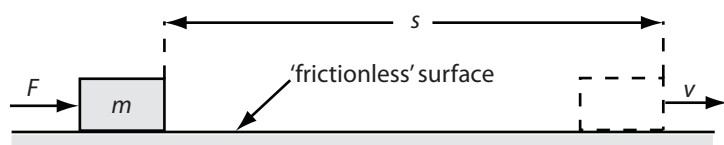


Figure 1

Analysing the data

Note: At this stage only the variables need to be used to obtain answers. That is, no numerical values have yet been assigned to the quantities.

- 1 Express the acceleration of the mass over the distance s using Newton's second law.

- 2 Starting with the formula $W = F \times d$, determine the work done (W) by the force in terms of m , a and s .

- 3 This amount of work is accounted for by recognising that the mass now has energy due to its motion, something that it did not have before, i.e.:
Work done = Gain in energy
and this energy is called 'kinetic energy' (KE). Express this kinetic energy in terms of m , a and s .

- 4 The equation of motion $v^2 = u^2 + 2as$ is now to be used to eliminate a and s in the equation derived in question 3. The final equation should link KE with m and v , the final velocity. Carry out this elimination and write the new equation for KE.

- 5 Using SI units and given the additional data that $m = 2$ units, $s = 4$ units and $F = 20$ units, calculate the following values, giving the SI units for each:

a W _____ a _____
 v _____ KE _____

- b Has the work done been conserved as KE? (Yes/No)

Data: Transformation of PE to KE

Consider an object of mass m which has been raised to a height h above ground level where it is motionless. It is then dropped and allowed to accelerate by g until it strikes the ground (Figure 2). The types of energy it possesses change, but its total energy remains a constant. This can be checked as follows.

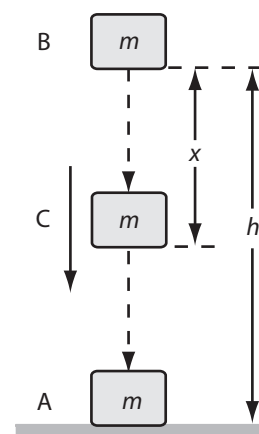


Figure 2

Analysing the data

Note: Again, only the variables need be used.

- 6 For the object, determine:
 - a PE. _____
 - b KE at B. _____
 - c total energy. _____
- 7 What force acts on the mass? _____
- 8 What work is done by the gravitational force?

- 9 Determine the velocity, v_f , of the object as it reaches point A. _____
- 10 Determine the KE of the object as it reaches point A.

- 11 What is the total energy of the object at the instant it reaches point A? _____
- 12 At the instant the object is at C, determine its:

PE. _____	velocity, v_C . _____
KE. _____	total energy. _____

Interpreting the data

- 13 What is true about the total energy possessed by the object at each of the three points considered?

- 14 Do the findings in this analysis agree or disagree with the Law of Conservation of Energy? _____

INTRODUCTION

To make an aircraft take off, fly at high altitudes and land safely involves a huge number of calculations and quite a lot of physics. This is equally true for a large commercial passenger aircraft and a small aircraft.

Aim

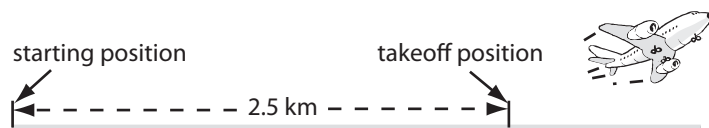
To compare some of the physical data relating to two such aircraft.

Note: The data supplied applies to straightforward situations and does not relate to such issues as strong head or tail winds, etc.

Data: A large commercial jet aircraft

The data items are approximate, but close to the real situation.

- The runway is required to be a minimum of 3.3 km long.
- The aircraft has four engines, each capable of a maximum thrust force of 244 kN.
- The mass of the empty aircraft (no fuel or passengers) is 178×10^3 kg.
- The maximum takeoff mass is 378×10^3 kg.
- Fuel capacity is 183×10^3 L.
- Density of aviation fuel can be taken as 0.75 kg L^{-1} .
- Acceleration due to gravity = 10 m s^{-2} .



Analysing the data

Note: 'Rounding off' procedures with large numbers may result in answers slightly different from those supplied.

- With an initial velocity of 0 and a final (takeoff) velocity of 200 km h^{-1} being achieved over 2.5 km, determine the average acceleration of the aircraft down the runway.
- What thrust force must be applied by the engines to make the aircraft, with maximum takeoff load, achieve the acceleration from question 1?
- What upwards force is needed to actually lift the aircraft from the ground as it takes off?
- Suppose that the aircraft being considered achieves an altitude of 10 000 m and then maintains a steady cruising speed. The air pressure at this altitude is

approximately 26 kPa (kilopascals) but it is 101 kPa at takeoff.

- What effect would the high altitude and low pressure have on the uplift force on the aircraft?
 - How can the necessary uplift be maintained in the thinner air?
- What is the momentum of the aircraft, fully laden, if the cruising speed is 900 km h^{-1} ?

Data: A small aircraft

A small aircraft has a laden mass (pilot, passengers and fuel) of 975 kg for takeoff. Its single engine has an output of 113 kW. It can take off in approximately 600 m and its takeoff speed is 120 km h^{-1} .

Analysing the data

- Determine the average acceleration of the aircraft down the runway.
- Calculate the force required to produce the acceleration in question 6.
- For what time is the force in question 7 applied?
- What work is done by the engine in accelerating the plane down the runway?
- Determine the power output of the single engine as it works to accelerate the aircraft.
- How does this power output compare to the claimed maximum power output of 113 kW?

Interpreting the data

- Which aircraft has the greater acceleration on the runway?
- All aircraft have maximum landing masses that are much less than their maximum takeoff masses. What factor would automatically make an aircraft's landing mass less than its takeoff mass?

INTRODUCTION

The term 'torque' is one that is often used in relation to *forces* and their effects. Another term that means the same is 'moment'. The torque or moment of a force is a measure of its turning effect about a point. The simple actions of turning a key in a lock, opening a door and turning a tap on or off all involve torque.

Aim

To understand the meaning of torque and look at some of its various applications.

Data: Force at an angle

The torque of a force depends upon (a) the *size* of the force and (b) the *perpendicular distance* from the line of action of the force to its *fulcrum* or *turning point*. Hence the formula for torque is:

$$\text{Torque} = \text{Force} \times \text{Distance (perpendicular)}$$

The symbol for torque is τ , and the SI units are newton metres (Nm). Three diagrams are supplied and you are asked to determine the torque in each case.

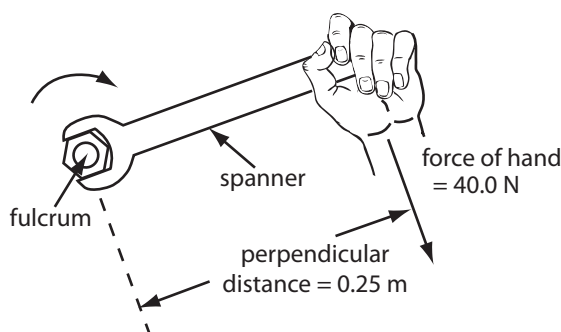


Figure 1

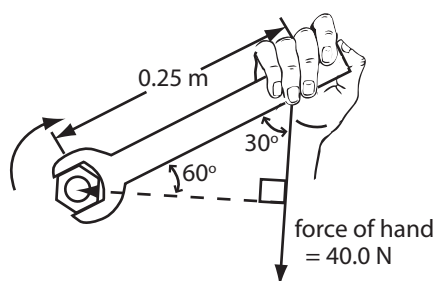


Figure 2

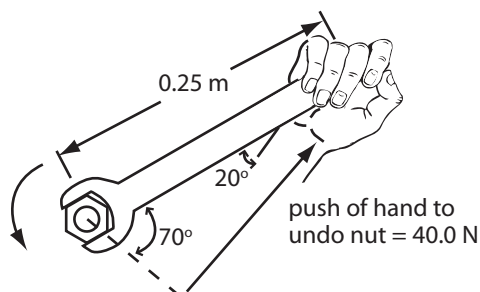


Figure 3

Analysing the data

- 1 Suggest a reason why the units of torque are written as Nm rather than mN.

- 2 Determine the torque of the force in each of the three situations.
Figure 1 _____
Figure 2 _____
Figure 3 _____
- 3 In Figure 3 the torque is applied to trying to 'undo' the nut. If the nut requires a torque of 4.0 Nm, is the torque supplied sufficient to do this? (Yes/No) _____

Data: Opposing forces

In some situations two torques oppose one another and may even cancel out each other's effects. A simple situation of this type is a see-saw. Here two students sit on opposite ends of the see-saw, creating opposing torques. Figure 4 shows the situation.

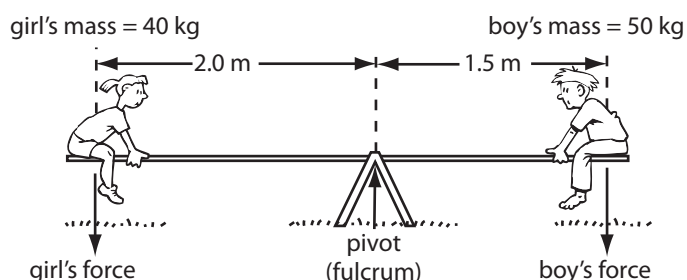


Figure 4

Analysing the data

- 4 a Taking a_g as equal to 10 m s^{-2} , determine the:
 - i boy's downward force. _____
 - ii boy's torque. _____
 - iii girl's downward force. _____
 - iv girl's torque. _____
- b Your answers should show that the two opposing torques are not quite equal in size. Which is greater?

- 5 To make the two torques equal in size, the fulcrum can be moved sideways. To accomplish this, in which direction would it have to be moved, towards the girl or towards the boy?

- 6 If the fulcrum cannot be moved to equalise the two torques and the boy decides to move to achieve equalisation, in which direction would he have to move?

Data: Addition of torques

In other situations two torques may combine if they act in the same rotational sense, e.g. both clockwise or both anticlockwise. In such a case the two forces are often the same size, constituting a 'couple'. This is shown in Figure 5, where $F_1 = F_2$ and both forces exert a clockwise torque about point P.

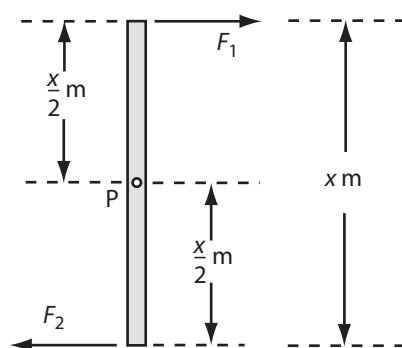


Figure 5

Analysing the data

- 7 If $F_1 = F_2 = 50 \text{ N}$ and $x = 3 \text{ m}$, determine the torque of each force and hence their combined turning effect.

Data: Torque of a couple

A specific example of the torque of a couple is the electric motor principle in which the forces produced on each side of a rectangular loop of wire carrying an electric current and in a magnetic field combine to turn the loop. Figure 6 shows a loop in such a situation and Figure 7 shows the same loop from the side. The force (F) on each side of the loop is given by the formula $F = BIl$, where B is the magnetic field strength, I is the current flowing through the loop and l is the length of the side of the loop. The left-hand side of the loop is pushed up while the right-hand side is pushed down, and so each force applies a torque and attempts to rotate the loop about its axis, XY.

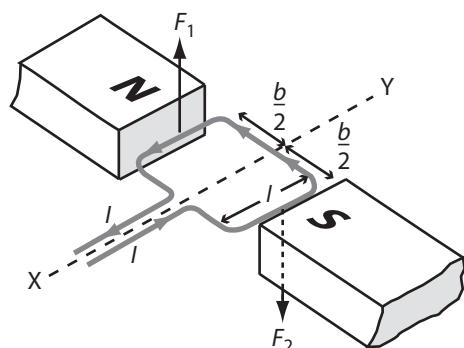


Figure 6

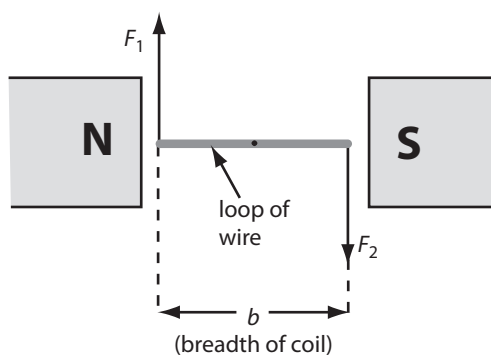


Figure 7

Analysing the data

- 8 Write an equation that could be used to determine the torque of each of F_1 and F_2 , using the data supplied.
- 9 Add the two torques τ_1 and τ_2 , which are equal in size, to give τ , their combined effect.
- 10 Explain why F_1 and F_2 are equal in size.
- 11 The 'formula' for τ derived in question 9 can be simplified by considering what length (l) $\times$ breadth (b) represents about the loop. Simplify the formula for τ , with this consideration in mind.

Interpreting the data

- 12 Supposing that, in a particular example, $B = 10 \text{ tesla}$, $I = 2 \text{ A}$, $l = 6 \text{ cm}$ and $b = 3 \text{ cm}$. Determine:
- the upward force F_1 .
 - the torque of F_1 .
 - the combined torque of F_1 and F_2 .
- 13 A second loop, identical to the first and insulated from it, with the same current passing through it, is now laid on top of the first loop. What might the total torque be now?

14 The work of Henry Cavendish

INTRODUCTION

The British scientist Henry Cavendish devised an ingenious method of measuring the gravitational force of attraction between two masses in the laboratory. Although he did not determine the value of G , the universal gravitational constant, his experiments and apparatus enabled others to do so. Cavendish was interested in determining the mean density of the Earth and he did this using his equipment.

Aim

To look at some of the experimental work done by Cavendish and its use in determining the value of G and some related physical values.

Data

Cavendish used a 'torsion balance', which consisted of two small lead spheres ($m = 0.73 \text{ kg}$) on the ends of a rod that was suspended from a long wire. As these spheres were brought near two large, massive spheres ($M = 158.00 \text{ kg}$), the attractive gravitational forces caused the rod to rotate and so twisted the wire. The extent of the twisting gave a measure of the very small attractive force between the small and large spheres. The principle of this balance is shown in Figure 1. The separation of the centres of the spheres (r) was approximately 0.405 m , as shown in Figure 2.

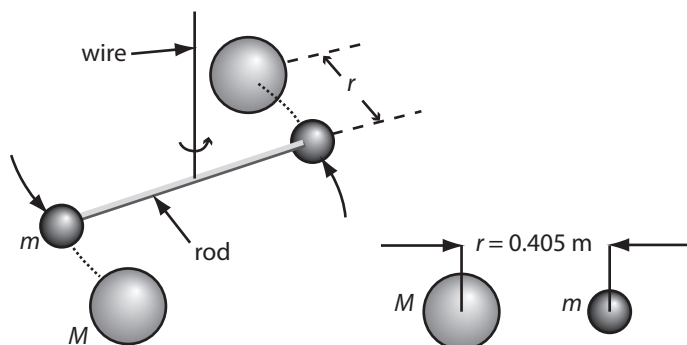


Figure 1

Figure 2

Analysing the data

- 1 Using the values of M , m and r above, and knowing that the modern value of G is accepted as 6.67×10^{-11} , determine the size of the attractive force F_a between the two spheres in the original experiment. Note: $F_a = GmM/r^2$.

- 2 Cavendish's calculated value of the mean density of the Earth was 5.45 kg m^{-3} . Today's accepted value is 5.52 kg m^{-3} . Considering these two values and the magnitude of the value of F_a in question 1,

comment on the sensitivity and accuracy of Cavendish's measurements.

Interpreting the data

- 3 Once G is known, the acceleration due to gravity a_g at the Earth's surface can also be determined, given that the mass of the Earth $M_E = 5.983 \times 10^{24} \text{ kg}$ and the radius of the Earth $R_E = 6.37 \times 10^6 \text{ m}$. Since $F_a = ma_g$ (from Newton's second law) and $F_a = GmM_E/R_E^2$ these two expressions can be equated. Do this and so derive a new equation with a_g the subject.

- 4 Substitute the values of G , M_E and R_E into the new equation and so calculate the value of a_g .

- 5 How does this value compare with the commonly accepted value of 9.8 m s^{-2} for a_g ?

- 6 Given that a_g is used in the equation $F = ma_g$, what units, other than m s^{-2} , are also appropriate for a_g ?

- 7 According to Cavendish, the mean density of the Earth was 5.4 kg m^{-3} . The rocks at the surface of the Earth are much less dense than this, about 2.7 kg m^{-3} . This means that Cavendish really discovered something about the rocks of the interior of the Earth. What was this?

- 8 In conducting his experiment and measuring such tiny forces of attraction, Cavendish took two basic precautions. (a) He encased his apparatus in a wooden box in a closed shed. (b) He then went outside the shed and observed what happened through telescopes through small holes in the shed wall. Suggest his probable reasons for these two precautions.

a

b

INTRODUCTION

When an object moves in a circular path its velocity is constantly changing. However, it may move with a constant speed, in which case it is said to undergo *uniform circular motion*. Newton's laws of motion still apply, as in all cases of motion.

Aim

To consider aspects of the motion of an object being whirled around in the horizontal plane at constant speed on the end of a string.

Data

The object moves as shown in Figure 1. The view is from above. The object completes two revolutions every second, at constant speed. The diagram shows the position of the object at time = 0.0 s and again after the object has moved through an angle of 45° .

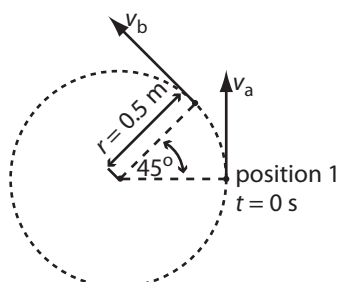


Figure 1

Interpreting the data

- 1 Although there has been no change in the speed of the object there has been a change in velocity. Explain how this has occurred. _____
- 2 The change in velocity, Δv , is linked to the initial velocity, v_a , and the final velocity, v_b , for the positions considered. Write an equation linking these three vector quantities. _____
- 3 Make Δv the subject of the equation for question 2. _____
- 4 Draw a vector diagram (to scale) in the space below, linking v_a , v_b and Δv as per the equation in question 3. Hence determine the magnitude of Δv . _____

- 5 In what direction does Δv appear to occur? _____
- 6 The change Δv over a given time means that there is acceleration. In what direction does this acceleration act? _____
- 7 Determine the value of the acceleration, given the data in Figure 1. _____
- 8 Calculate the force required to cause the acceleration in question 7 if the mass of the object is 0.1 kg. _____
- 9 What name is given to the force that causes uniform circular motion? _____
- 10 In this situation what supplies the force calculated in question 8? _____
- 11 If the speed of the object were to be increased, what would have to happen to:
 - a the acceleration? _____
 - b the force causing motion of the object? _____
- 12 What effect would the changes in question 11 have on the time of revolution of the object? _____
- 13 If the object moved in uniform circular motion for another revolution and the string broke as the object returned to position 1 ($t = 0$), it would no longer move along its original path. Which path, X or Y as shown in Figure 2, correctly shows how the object would continue to move after the string broke?

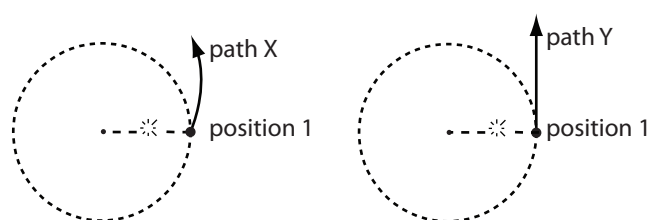


Figure 2

- 14 Explain why the object follows the path chosen in question 13. _____
- 15 Which of Newton's laws of motion explains the reason why the object moves along the path chosen in question 13? _____

INTRODUCTION

Until about 1880 'heat' was considered to be a weightless material fluid that flowed from one body to another. This fluid was called 'caloric'. The theory was useful in developing ideas about quantities such as heat and specific heat capacities. For example, the unit of heat was called the 'calorie', and was defined as that amount of heat that would raise the temperature of 1.0 g of water by 1.0 Celsius degree. However, evidence from further investigations suggested that this theory was not valid and that heat, like kinetic and potential energy, was a form of energy. James Joule reasoned that if heat was a form of energy, then it could be transformed into other forms and that there would be a definite 'conversion factor' according to the Law of Conservation of Energy. Joule set out to determine this conversion factor by changing mechanical energy into heat energy.

Aim

To collect some data from an experiment similar to Joule's experiment and so calculate the 'conversion factor'.

Data

Joule's apparatus is shown in Figures 1 and 2. A calorimeter was divided into compartments using baffle plates, and filled with a known mass of water. The solid masses M_1 and M_2 were wound up to the levels A and C respectively and then allowed to drop back through height h , to B and D. This caused the paddles in the calorimeter to spin and thus transformed the potential energy of the masses into heat energy of the water, so raising its temperature. This was repeated a number of times by winding the masses back up and repeating the drop. Any rise in the temperature of the water was recorded. Some data derived from such an experiment is given below.

- Mass of water in calorimeter = 4.00 kg
- Mass of M_1 = Mass of M_2 = 70.0 kg
- Number of times masses were dropped = 20
- Distance of drop = 3.0 m
- Temperature rise in water = 5.1°C

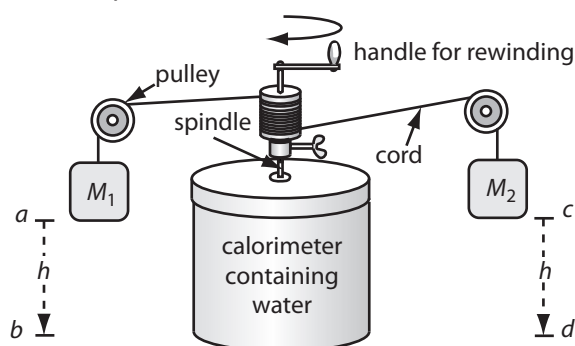


Figure 1

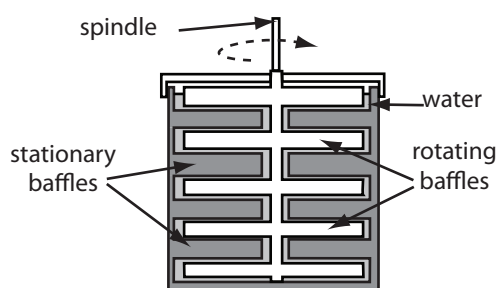


Figure 2 Interior of calorimeter

Analysing the data

- 1 Assuming a_g equals 9.8 m s^{-2} , calculate the loss of potential energy for each mass each time it was dropped.
- 2 Determine the total loss of potential energy of *both* masses after they were dropped 20 times.
- 3 Calculate the total number of calories required to raise the temperature of the water by 5.1°C.
- 4 Since the heat energy (calories) calculated in question 3 has come from the potential energy (joules) determined in question 2, the two quantities can be equated. Do this and then determine the number of joules required to produce 1.0 calorie of heat. This value is the 'mechanical equivalent of heat'.

Interpreting the data

- 5 The presently accepted value of the mechanical equivalent of heat is 4.18 joule = 1.0 calorie. How does this value compare with the value calculated in question 4?
- 6 Considering the apparatus used, suggest two possible sources of energy loss that might account for an experimental value being less than the accepted value.
- 7 Joule made provision for disconnecting the paddles from the drum as the masses were wound up. Why would this have been done?

INTRODUCTION

The term 'pressure' is a very commonly used one. Terms such as 'blood pressure', 'tyre pressure' and 'air pressure' are used on a daily basis. But just what does 'pressure' actually mean?

Aim

To investigate the meaning of the term 'pressure' and see how it applies to a few different situations.

Data: Set 1

Pressure is defined as the 'force per unit area'. Consider a rectangular brick of dimensions $25\text{ cm} \times 10\text{ cm} \times 8\text{ cm}$ with a mass of 2.0 kg . It can be put onto a planar surface in three different ways: **a** on its $25\text{ cm} \times 10\text{ cm}$ surface **b** on its $25\text{ cm} \times 8\text{ cm}$ surface **c** and on its $10\text{ cm} \times 8\text{ cm}$ surface. In each case the brick applies a force to the planar surface of $F = 2 \times 9.8\text{ N}$.

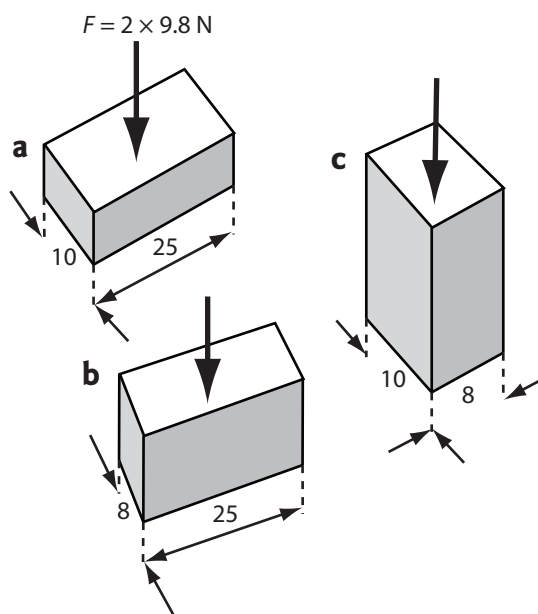


Figure 1

Analysing the data

- 1 Determine the surface area of each of the three different sized surfaces of the brick.

a _____
b _____
c _____

- 2 Convert these surface areas to m^2 .

a _____
b _____
c _____

- 3 Keeping in mind that the weight force of the brick is constant, calculate the pressure, in N m^{-2} , applied in each of the three situations when the brick is put onto a planar surface.

a _____
b _____
c _____

- 4 The pressure unit of N m^{-2} is called the 'pascal', i.e. $1\text{ N m}^{-2} = 1\text{ Pa}$. If the brick is held so that it rests on a corner (of area $= 1.0\text{ mm}^2$), determine the pressure, in pascals, that it exerts on a surface.

- 5 How does this compare with normal air pressure at sea level of approximately 101.3 kPa ?

Data: Set 2

The pressure exerted by the air changes with altitude. The table below shows some approximate values at the altitudes listed.

Altitude (m)	Pressure (kPa)
2 000	77
4 000	63
6 000	47
8 000	34
10 000	24
12 000	19
14 000	14



- 6 On the axes provided, construct a graph of air pressure vs altitude, remembering that the air pressure at sea level (altitude $= 0$) is 101.3 kPa .

Interpreting the data

- 7 Does the graph show that the air pressure changes uniformly with altitude? (Yes/No) _____
- 8 If an aircraft at sea level took off and ascended to $10\,000\text{ m}$, how many times greater would the air pressure inside the aircraft be than the air pressure outside? (approx.)

- 9 Would the difference between the inside and outside pressure tend to make the aircraft explode or implode?

INTRODUCTION

The pressure exerted by a liquid depends upon the density of the liquid and the depth into the liquid (i.e. the height of liquid above the reference point). The unit is the pascal and the formula for calculating pressure in liquids is $\text{Pressure} = \text{Height (m)} \times \text{Density (kg m}^{-3}\text{)} \times \text{Acceleration due to gravity (m s}^{-2}\text{)}$. Note: This is equivalent to the formula $\text{Pressure} = \text{Force/Area}$.

Aim

To investigate some aspects of pressure in water.

Data

The pressure in water increases quite markedly with depth. Since the density of pure water is 10^3 kg m^{-3} , this means that for every metre of depth into water, there is an additional 1000 kg above every square metre. Figure 1 shows this. If a_g is taken as approximately 10 m s^{-2} then the weight force due to this mass of water is $1000 \times 10 \text{ N} = 10^4 \text{ N}$. It should be noted that the force acting on the surface of the water is an additional (approximate) 100 000 Pa due to the air pressure.

Analysing the data

- 1 The formula given in the introduction applies to all fluids, including air. Given that the air pressure is approximately 100 000 Pa at the Earth's surface, what mass of air must be above 1 m^2 of the Earth's surface?

- 2 What would be the pressure due to the water only at a depth of 1.0 m? (This can be rounded to the nearest 1000 Pa.)

- 3 At the depth mentioned in question 2, what would be the total pressure due to the combined effects of the water and the air above it?

- 4 The pressure due to the air at the Earth's surface ($\approx 100\,000 \text{ Pa}$) is often used as a standard and given the value of 1.0 atmosphere (atm). What depth of water (alone) would produce a pressure of 1.0 atm?

- 5 In the sport of 'free diving' a diver takes one breath and then, standing on a platform connected to a pole and weights, is plunged to a depth from which he/she must return on the single breath. A world record was set in the Caribbean when a woman 'free dived' and was plummeted to a depth of 160 m to break

both the men's and women's world records. Calculate the maximum pressure, in atmospheres, to which her body was subjected.

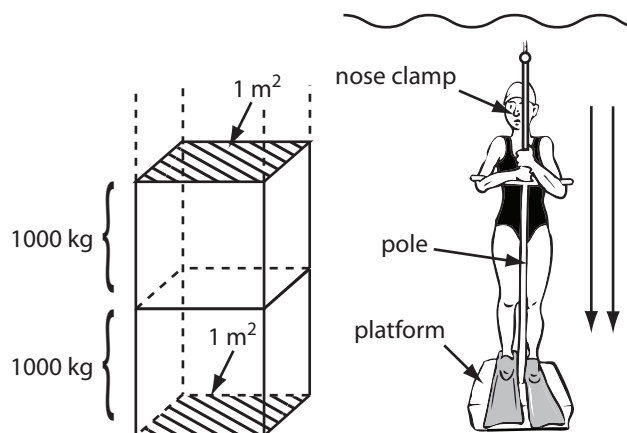


Figure 1

Figure 2

- 6 Normally, the volume of a set mass of air depends upon the pressure acting on it, and the product of $\text{Pressure} \times \text{Volume}$ is a constant; that is, if the pressure is doubled, then the volume is halved. If a balloon containing 3.4 L of air at 1.0 atm was taken to the same depth as the free diver in question 5, what would be its new volume?

- 7 If the diver in question 5 could hold her breath for, say, 4.0 minutes, what would be her average speed (m s^{-1}) during the combined descent and ascent?

- 8 The calculations above have considered the density of the water to be $1.0 \times 10^3 \text{ kg m}^{-3}$. If ocean water has an average density of $1.02 \times 10^3 \text{ kg m}^{-3}$, how would this more realistic value affect the calculated pressure at the depth of 160 m achieved by the diver?

- 9 The density of ocean water is always greater than that of pure water. Suggest a reason for this.

- 10 An extremely dense liquid such as mercury, which is 13.6 times as dense as pure water, would create great pressures at even modest depths. At what depth of mercury would a pressure equal to that acting on the free diver at 160 m be achieved?

INTRODUCTION

When two electric charges are brought near one another they experience a force. If the two charges are of similar sign (i.e. both positive or both negative), the force is a repulsive force.

Aim

To use the results of experiments to try to find out if a relationship exists between the size of the force on the spheres and the distance between them.

Data

An experiment was carried out using two light, small styrofoam spheres supported by long threads of fine silk. Identical charges were placed on the spheres, which then repelled one another. Restoring forces were applied to bring the spheres back to their original positions. This enabled the repulsive force between the charges to be measured. The distance (d) was varied, as shown in the table, and in each case the repulsive force (F) was determined. The experiment was repeated a number of times to calculate average readings. These readings are listed in the table and the apparatus is shown in Figure 1.

$d (\times 10^{-3}) \text{ m}$	$F (10^1) \text{ N}$	x	y
20.0	9.00		
30.0	4.00		
40.0	2.25		
50.0	1.44		
60.0	1.00		
70.0	0.73		

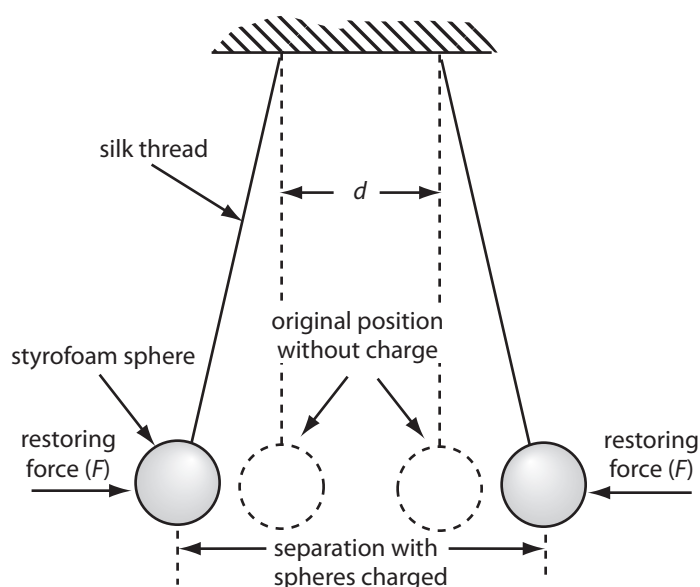
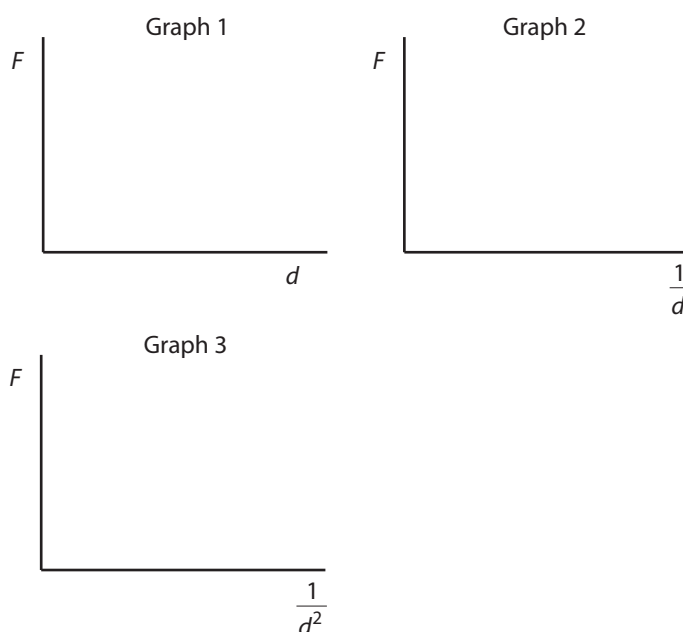


Figure 1

Analysing the data

- To find what relationship may exist between F and d , graphs need to be constructed. Calculate the values of $\frac{1}{d}$ and $\frac{1}{d^2}$ and write these values into columns x and y respectively. There is no need for detailed units here, but consistency with orders of magnitude is always essential. Plot the appropriate data on the axes below and so determine the relationship that exists between d and F .



Interpreting the data

- Which graph shows a direct relationship between the two variables?

- Write, in mathematical form, the relationship between d and F .

- Explain why styrofoam spheres and silk thread were used.

- If one of the charged spheres had been given a positive charge and the other a negative charge, would this have changed the shapes of the graphs? (Yes/No)
_____ Explain. _____

INTRODUCTION

A method of determining the value of an unknown resistance is an electrical set-up called the 'Wheatstone bridge', named after its inventor. It measures the value of resistance more accurately than most other methods.

Aim

To use the Wheatstone bridge method to determine the value of an unknown resistance. This exercise assumes a basic knowledge of Ohm's law.

Data

The Wheatstone bridge is an arrangement as shown in Figure 1 in which four resistors are connected in a circuit with a source of potential difference and a sensitive galvanometer across the points C and D. Usually three resistors are of known value and one (R_1 in this case) is the unknown. The resistors are selected with resistance values such that, when the switch is closed, the reading on the galvanometer is zero; that is, there is no potential difference between the points C and D. When the switch is closed and a current I flows, the current is divided at A through the two pathways to B; that is, it splits into I_1 and I_2 as shown. As the current passes through the resistors there is a potential drop across each one.

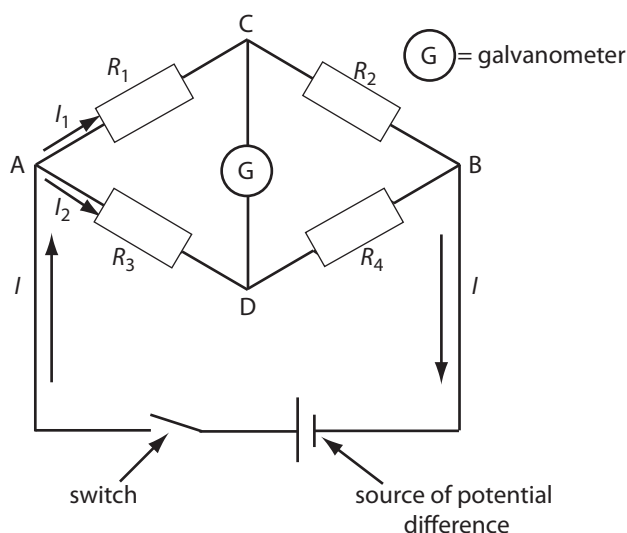


Figure 1

Analysing the data

- 1 The points A, B and C are at different potentials. Arrange them in order from highest to lowest potential.

- 2 Arrange points A, B and D in order from highest to lowest potential.

- 3 Since there is no potential difference between C and D, what must be true about the potential difference across AC and AD?

Interpreting the data

- 4 Using Ohm's law, express the potential difference across AC in terms of the current travelling through it and its resistance. Repeat this for the pathway AD.
a AC _____ b AD _____
- 5 The two potential differences in question 4 are equal and so can be equated. Write down the resulting equation (equation 1).

- 6 A similar logic applies to the pathways CB and DB. Again, use Ohm's law to express the potential difference across these pathways in terms of the current and the resistance.
a CB _____ b DB _____
- 7 The two expressions in question 6 can be equated, since they represent the same potential difference. Write down the equation (equation 2).

- 8 Now divide equation 1 by equation 2 and write the new equation (equation 3) in its simplest form.

- 9 Rearrange equation 3 to make R_1 the subject.

- 10 This final equation becomes a formula by which the unknown resistance can be determined if the other three resistances are known. It is the basic formula for using a Wheatstone bridge. Does it depend on the potential difference applied to the circuit? (Yes/No)

Using the formula

- 11 Now that a formula has been established, it can be used to determine the value of an unknown resistance R_1 . Suppose that, in such an experiment, when the galvanometer reading was zero the values of the other resistors were $R_2 = 4.0$ ohms, $R_3 = 6.0$ ohms and $R_4 = 8.0$ ohms. What must be the value of R_1 ? _____
- 12 What is the ratio of $R_3 : R_4$? _____
- 13 Suppose that resistors R_3 and R_4 were replaced with resistors of values 12.0 ohms and 16.0 ohms respectively.
 - a Would this alter the calculated value of R_1 ? (Yes/No)
 - b What would be the ratio of $R_3 : R_4$ in this case?

 - c How does this new ratio compare to the ratio in question 12? _____
- 14 The answer for question 13 leads to the fact that the exact values of R_3 and R_4 need not be known. In that case, what must be known about them instead?

INTRODUCTION

Transformers are electrical devices that are designed to change electrical potential (voltage) as desired. They can be made to increase a voltage (step-up) or decrease a voltage (step-down).

Aim

To investigate the use of transformers in four situations that relate to aspects of daily living.

- 1 An LCD TV operates on a normal household input voltage of 240 V. However, its transformer lowers the voltage to the TV to 15.0 V.
- 2 A laptop computer also plugs into the 240 V household circuit, but its transformer changes the voltage input to the computer to 19.0 V.
- 3 An electricity generator in a power station has an output voltage of 2.00×10^4 V. This voltage is fed into a transformer which changes it to 5.00×10^4 V before it is transmitted through the high-voltage wires to the city.
- 4 A handheld electric light that plugs into a 240 V household circuit has a transformer that reduces its voltage to 32.0 V.

The table below summarises the facts about the four situations with a little more data.

Interpreting the data

- 1 The power of an electric circuit is calculated by the following relationship:

$$\text{Power} = \text{Voltage} \times \text{Current}$$

For each of the four devices, calculate the input power and the output power and enter the values in the table.

- 2 Determining the efficiency of a device is a way of comparing the rate of energy input with the rate of energy output. It is calculated by the following formula:

$$\text{Efficiency} = \frac{\text{Power output}}{\text{Power input}} \times 100$$

Determine the efficiency of each of the four devices and enter the values in the table.

- 3 The energy loss seems to be least for the generator transmitting electricity to the home. Suggest why this is the case.

- 4 Which of the situations involve:

a a step-up transformer?

b a step-down transformer?

- 5 In each situation there appears to be a loss of energy during transformation. Suggest how this energy can be accounted for.

- 6 Situations 1, 2 and 4 involve low output voltages for devices that are used in or around the home. What might be the reason for this?

- 7 The laptop computer is used for 0.5 h on the household 240 V supply.

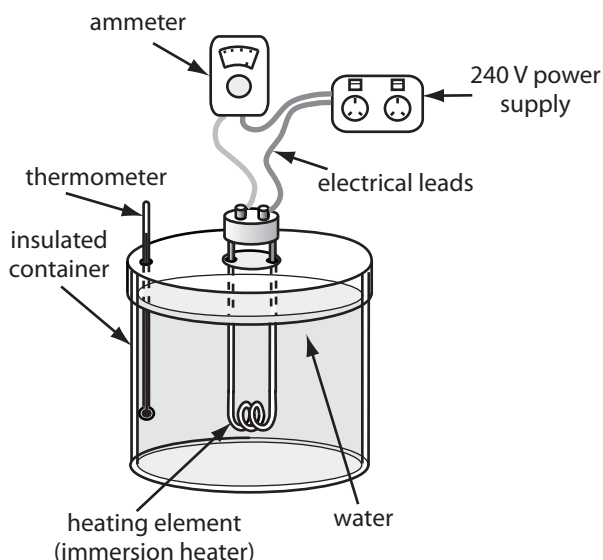
a Calculate the total amount of energy consumed in joules (J).

b Determine the amount of energy (J) wasted or not converted into electrical energy by the transformer.

Situation	Input voltage (V)	Input current (A)	Output voltage (V)	Output current (A)	Input power (W)	Output power (W)	Efficiency $\frac{P_o}{P_i} \times 100$ (%)
LCD TV	240	0.29	15.0	4.50			
Laptop	240	1.50	19.0	3.42			
Power station	2.00×10^4	100.00	5.00×10^4	39.60			
Light	240	0.6	32.0	4.00			

INTRODUCTION

The mechanical equivalent of heat (Joule's equivalent) can be determined using an electrical heating method in which an immersion heater is placed into a known mass of water at a measured temperature. The electrical energy in the heating element is transferred to the water, causing a temperature rise. The amount of energy can be determined from both the electrical data and the rise in temperature of the water. Hence a 'conversion factor' can be determined. A simplified version of the experimental apparatus is shown in the diagram.



Aim

To determine such a conversion factor from experimental data and to use the factor in further calculations.

Data

Some data from the experiment:

- Mass of water used = 1.6 kg
- Initial temperature of water = 14.3°C (K)
- Final temperature of water = 100.0°C (K)
- Voltage supplied to circuit = 240 V
- Current passing through heating element = 5.0 A
- Time of current flow = 8 min

Analysing the data

- 1 Determine the rise in temperature of the water being heated.

- 2 Determine the amount of electrical energy supplied by the heating element during the 8 minutes, given that $P(W) = \frac{E(J)}{t(s)} = PD(V) \times I(A)$. Give the answer in kilojoules. (PD = potential difference)

- 3 Assuming that all of the energy in question 2 goes into the water, use the following equation to determine the 'conversion factor'. Note: Be careful with indices! Make sure that the units are $\text{kJ kg}^{-1} \text{K}^{-1}$.

$$\text{Energy supplied to water (kJ)} = \text{Mass of water (kg)} \times \text{Conversion factor} \times \text{Temperature rise (K or } ^\circ\text{C)}$$

Interpreting the data

Once the conversion factor has been calculated as above, it can be used to determine other aspects of heating by this method. Three examples are given below.

- 4 An electrical hot water system operates at 240 V and holds 300 L (= 300 kg) of water. Cold water is fed into the heater at 15°C and heated to 50°C by a current of 10 A. How long would it take to get the water to the desired temperature?

- 5 An electric kettle is listed as '240 V, 1200 W'. Water (1.5 L) is heated in it for 7 minutes. Is the energy supplied enough to make the water boil; that is, reach 100°C if its initial temperature is 15°C? Show your working.

- 6 The cigarette lighter plug in a motor vehicle is often used as an outlet which will supply 12 V to appliances that can be used in the vehicle. One such appliance is a small electric jug that is able to bring 300 mL (= 0.3 kg) of water at 20°C to the boil in 8.0 minutes.

- a Determine the amount of heat energy required to boil the water.

- b Determine the current supplied by the car's battery to produce the required energy.

INTRODUCTION

Two basic types of mechanical wave transmission that commonly occur are (a) transverse and (b) longitudinal. Both types are transmitted in elastic media. The longitudinal mechanical waves are transmitted because the medium shows some resistance to compression forces and so 'rebounds' in an elastic manner. The speed of the rebound is reflected in the 'compression modulus' for the medium. The bigger the value of this modulus, the more elastic the medium is to compression waves and the greater the speed of the waves in it. A similar logic applies to mechanical transverse waves in media that are subject to shear forces by the waves and so each medium is assigned a 'shear modulus'.

Aim

To investigate aspects of mechanical shear (i.e. transverse) waves and their transmission through various media.

Data

The speed of such waves in a specific medium is determined from the formula

$$v \text{ (speed)} = \sqrt{\frac{G}{\rho}}$$

where G is the shear modulus and ρ is the density of the medium.

The shear modulus and density of each of eight media are listed in the table below. The units are consistent but can be ignored for the purpose of the exercise.

Medium	Shear modulus (G)	Density (ρ)	Speed (v)
a Diamond	478.0	3.51	
b Steel	79.3	7.86	
c Copper	44.7	8.92	
d Titanium	41.4	4.54	
e Glass	26.2	2.65	
f Aluminium	25.5	2.70	
g Polythene	0.117	0.92	
h Rubber (butadiene)	0.0006	0.62	

Analysing the data

- 1** According to the formula given, what happens to the speed (v) of shear waves as the density of the medium increases?

- 2** Which medium in the table has:
- the least resistance to shear forces?

 - the most resistance to shear forces?

- 3** Which medium would probably have the lowest compression modulus?

- 4** Calculate the speed of shear waves in each medium, using the formula provided, and enter the values in the table.

- 5** Does the completed table now show that:
- as the density of a medium increases the speed of the shear waves also always increases? (Yes/No)

 - as the shear modulus of a medium decreases the speed of the shear waves also always decreases? (Yes/No)

Interpreting the data

- 6** Water is a medium that is usually not listed in shear modulus tables. Suggest a reason for this.

- 7** Light and other electromagnetic wave motions are transverse waves and do not follow the 'rules' of the waves dealt with here. Suggest a reason for this.

INTRODUCTION

The behaviour of both mechanical transverse (or shear) and compression waves in rock materials obeys the same rules as when those waves travel through substances such as iron, copper or concrete.

- 1 As the density of the material increases, the speed of such waves decreases.
- 2 Both types of waves increase in speed as the modulus of shear ($\approx$ resistance to shear forces) and the modulus of compression ($\approx$ resistance to compression) increase respectively.

Aim

To investigate the behaviour of some naturally occurring mechanical transverse and compression waves as they travel through the Earth.

General data

When an earthquake occurs in rocks beneath the Earth's surface (focus), it produces both transverse (shear or S) and compression (P) waves, which travel outwards in all directions from the focus into the surrounding rocks, as shown in Figure 1.

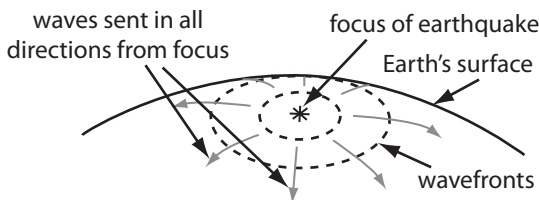


Figure 1

Data

- 1 A layer of three different substances with similar moduli of compression but with increasing densities are placed on top of one another. P waves are transmitted into the top layer and progress through the layers as shown in Figure 2.

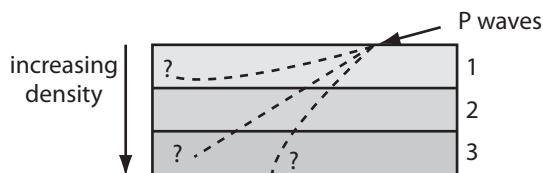


Figure 2

- 2 A different set of three substances, this time with similar densities but with increasing moduli of compression are placed on top of one another and, again, P waves are transmitted into the top layer as shown in Figure 3.

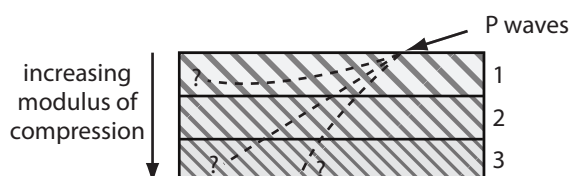


Figure 3

- 3 As P waves from an earthquake move into the mantle of the Earth beneath the crust, their speeds increase with depth and the waves are refracted, as shown in Figure 4, even though the density of the rocks of the mantle increases steadily. In addition, the pathways of these waves as they pass into the core of the Earth undergo a dramatic change in direction at point X, as shown in Figure 5.

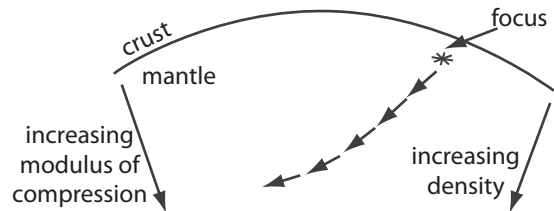


Figure 4

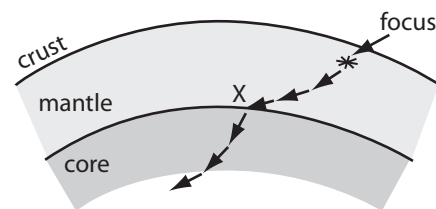


Figure 5

Analysing the data

- 1 Complete the diagrams for the two situations in Figures 2 and 3, showing the approximate, expected refracted pathways.
- 2 Which property of the rocks – density or modulus of compression – must be the dominant one to produce the refraction? _____

Explain. _____

- 3 What change in speed of the waves would accompany this directional change at point X in Figure 5? _____

- 4 What does the refraction shown at point X in Figure 5 suggest about how the material of which the core is composed compares with that of the mantle? _____

- 5 An added piece of information is that S waves reaching the core cannot to be transmitted through it. Instead, they appear to be absorbed. What does this suggest about the modulus of shear of at least part of the core? _____

- 6 Make a general statement about the structure of the interior of the Earth as revealed by this earthquake data. _____

INTRODUCTION

Optical fibres are used increasingly in communications and medical procedures.

Aim

To understand some principles used in these fibres which enables them to transmit wave motions as they do.

Data

The phenomenon of total internal reflection occurs in some situations when light travelling in one medium strikes a boundary with a second medium of lower refractive index. If the angle of incidence (i) is small, refraction occurs as the light pass into the second medium (Figure 1). As the angle of incidence is increased, a point is reached at which the refracted ray passes along the surface of separation of the two media (Figure 2). If the angle of incidence is increased beyond this point, the light cannot leave the original medium. Instead, it is totally reflected within this medium, with the reflection obeying the normal laws of reflection (Figure 3). The angle of incidence that results in refraction along the surface of separation is called the critical angle (θ_c). It is related to the refractive index (RI) of the original medium by the relationship:

$$\sin \theta_c = \frac{1}{\text{RI}}$$

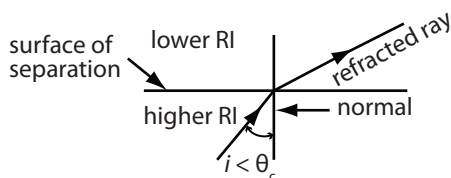


Figure 1

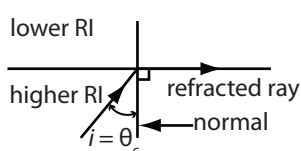


Figure 2

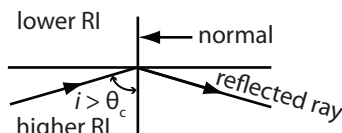


Figure 3

The following table shows either the RI or the θ_c for a number of transparent media when each is surrounded by air, which is given a RI of 1.00.

Substance	RI	θ_c
Water		48.80°
Flexible glass	1.46	
Plastic 1	1.44	
Plastic 2		41.14°
Plastic 3		42.16°

Analysing the data

- Complete the table, with the assistance of the formula supplied in the data.
- Optical fibres have to be solid, flexible substances such as the three varieties of plastic listed. Usually, a plastic optical fibre is encased in another plastic to protect it. The light passing through the inner fibre must not pass out of it. For each of the combinations below, state whether or not the combination would be suitable for an optical fibre.

a Flexible glass inside, plastic 3 outside

b Plastic 2 inside, flexible glass outside

c Plastic 1 inside, plastic 2 outside

d Plastic 3 inside, plastic 1 outside

Interpreting the data

- There are other combinations of the four substances that could be used to make optical fibres. Give three such examples, indicating the substance that must be on the inside by naming it first.
 -
 -
 -
- Which pair of substances would produce a combination with:
 - the biggest θ_c ?
 - the smallest θ_c ?
- The light used to send signals down optical fibres is usually laser light, rather than normal light. Suggest a reason for this preference.
- Once a laser signal is sent down an optical fibre it is detected at the other end and converted into another form of energy that is then fed into a computer for 'decoding'. Suggest what type of energy the laser light is converted into.

INTRODUCTION

When a string (including a fine wire) is held at two points with a fair degree of tension on it and then 'plucked', it vibrates to and fro with a particular frequency. The frequency produced depends upon a number of factors such as (1) the length, (2) the tension and (3) the mass per unit length of the string. These three factors affect the string differently and act together to determine the final frequency of the vibration.

Aim

To use data to try to find relationships between the three factors mentioned and the frequency.

Data

Three separate experiments were carried out. The results of each of these experiments and the variables investigated are shown below. The units are not listed, but there is consistency in each set of results.

Experiment 1

In this experiment the *tension* in the string was constant and the *same string* was used each time. The length of the string was varied and the resulting frequency was determined in each case.

Length (l)	Frequency (F)	$\frac{1}{l}$	$F \times l$
100.0	256.0		
50.0	512.0		
25.0	1024.0		
12.5	2048.0		

Experiment 2

In this experiment the length of the string was constant and the same string was used each time. The tension applied to the string was varied and the resulting frequency was measured in each case.

Tension (T)	Frequency (F)	$\sqrt{T}$	$F\sqrt{T}$
25.0	100.0		
100.0	200.0		
400.0	400.0		
625.0	500.0		

Experiment 3

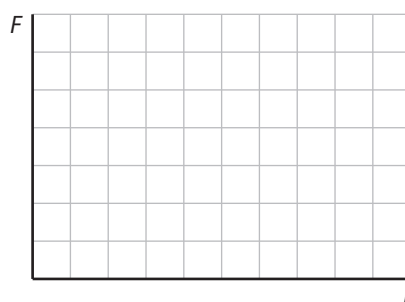
The length and the tension were kept constant, but strings of different mass per unit length (m) were used. The frequency was measured for each string used.

Mass per unit length (m)	Frequency (F)	$\sqrt{m}$	$\frac{1}{\sqrt{m}}$	$F \times \sqrt{m}$
0.01	100.0			
0.04	50.0			
0.09	33.3			
0.16	25.0			

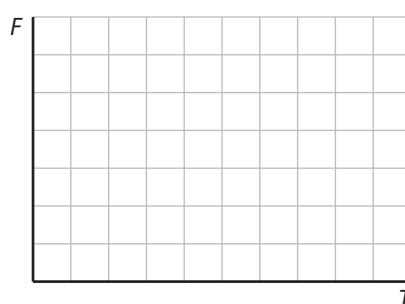
Analysing the data

- I For each set of data draw a graph showing the effect of the variable quantity (x-axis) on the frequency (y-axis).

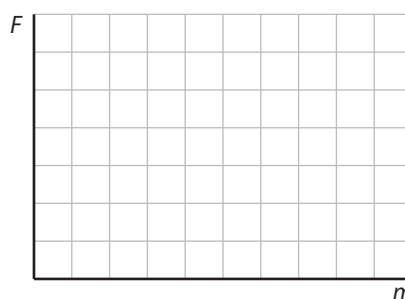
Experiment 1



Experiment 2

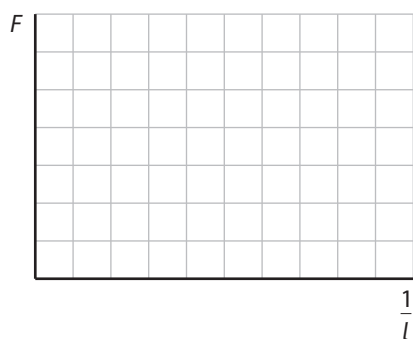


Experiment 3

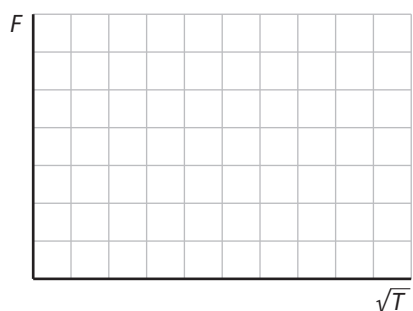


- 2 a** Are any of these graphs straight lines? (Yes/No) _____
- b** Do any of the graphs show a direct relationship between the variables? (Yes/No) _____
- 3** In order to search for any direct relationships it is necessary to calculate values to complete the tables. Complete the tables and draw these graphs on the axes provided below.

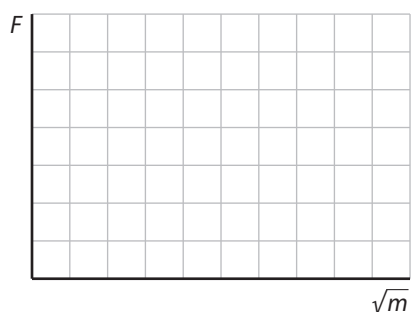
Experiment 1



Experiment 2



Experiment 3

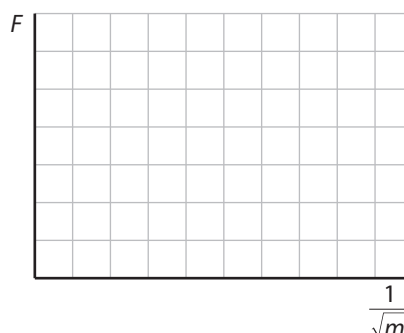


- 4 a** Are any of these graphs straight lines? (Yes/No) _____
- b** If so, which one(s)? _____

- 5** For the values of the data in Experiment 3, one more attempt at a graph may need to be made.

Plot the values of frequency (y-axis) against

$$\frac{1}{\sqrt{\text{mass per unit length}}} \text{ (x-axis).}$$



- 6** Does this graph plot as a straight line? (Yes/No) _____
- 7** Does the relationship between the two variables appear to be a direct one? (Yes/No) _____

Interpreting the data

- 8** At this stage the graphical processing of the data should have revealed that the frequency of a vibrating string depends upon three factors; that is, that the frequency is directly proportional to three variables. Write separate expressions to show these relationships.
- a** $F \propto$ _____
- b** $F \propto$ _____
- c** $F \propto$ _____
- 9** It should be noted that the last column of each table has a set of four values that are constant for that table. Do these values support the relationships expressed in question 8? (Yes/No) _____
- 10** As the frequency of a sound determines the pitch that we perceive (i.e. high frequency = high pitch), what effect would shortening the length of a violin string have on the pitch of the sound that we hear? _____

27 Water temperature change

INTRODUCTION

When heat energy is put into or taken out of a substance, there is usually a change in the size or volume of that substance. The amount of change depends on a number of factors including the physical state and nature of the substance and the extent of the change in temperature. A change in size means a change in density, since the mass of the substance is unchanged.

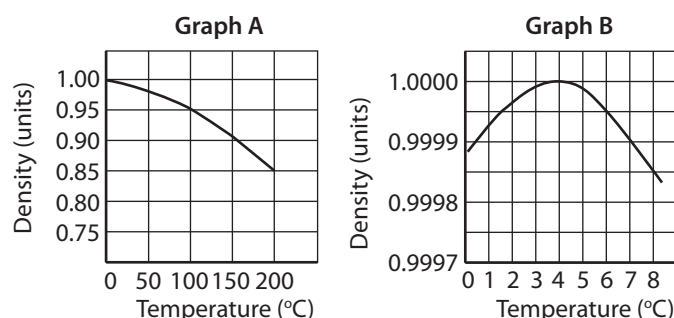
Aim

To study the change in density of a fixed mass of water as it is made to undergo temperature changes, and to interpret the effect of such changes.

Data

Two graphs are shown below, each relating the density of liquid water to its temperature. Graph A covers the large temperature range of 0–200°C; graph B only covers the small temperature range of 0–8°C.

Remember: $\text{Density} = \frac{\text{mass}}{\text{volume}}$.



Analysing the data

- 1 In graph A the water has been kept as a liquid to 200°C, although it normally boils at 100°C. Suggest how this might have been achieved.

- 2 Why is the 'upward bump' in graph B not obvious in graph A?

Interpreting the data

- 3 From the graphs predict:

a the density of water at 100°C.

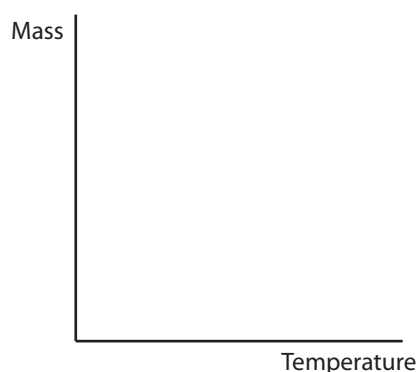
b two temperatures at which the density of water is 0.9999 units.

- 4 For graph A, make a statement that expresses the findings about the relationship of the density of water to its temperature.

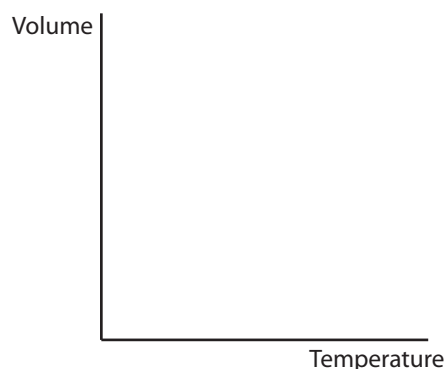
- 5 Graph A shows the trend that commonly applies to solids, liquids and gases as they are heated. What is the cause of this change as temperature increases?

- 6 Using graph A as a basis, plus any necessary data given, draw an approximate line graph on the axes provided to show any change(s) that may occur in the properties plotted.

Graph C



Graph D



- 7 Explain the reasons for choosing the shapes of graphs C and D.

Graph C _____

Graph D _____

- 8 In a cold climate the temperature may drop from 8°C to 0°C overnight. If a bottle of water is filled and sealed when its temperature is 4°C and left overnight, explain the forces that would occur on the bottle as its temperature dropped to 0°C.

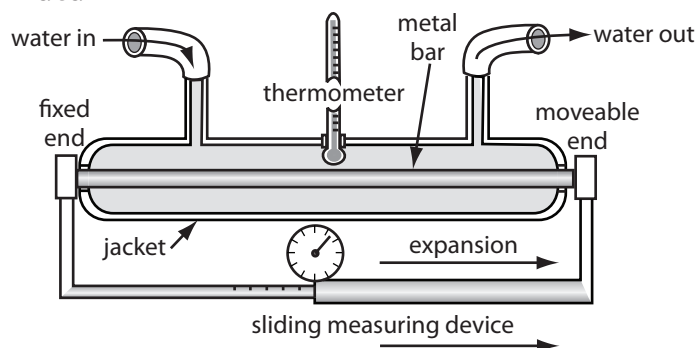
INTRODUCTION

It is well recognised that, when heated, metals expand or increase in size more than most non-metallic substances. Also, metals expand differently from one another so that a 1.0 m length of aluminium does not expand by the same amount as a 1.0 m length of iron when both metals are heated by the same amount.

Aim

To compare the relative amounts of expansion of different metals that are heated over the same temperature range.

Data



A bar of metal is set up inside a 'jacket'. Its length initially is 1.000 00 m and its temperature is 20°C. Water of increasing temperature is run through the jacket and a very sensitive measuring device records the length of the bar at specific temperatures. This experiment was repeated using three different metals: iron, aluminium and copper. The results of the experiment are shown in the table below.

Metal	Iron	Aluminium	Copper
Length at 20°C (m)	1.000 00	1.000 00	1.000 00
Length at 70°C (m)			
Length at 100°C (m)	1.000 96	1.002 08	1.001 36
Total increase in length (m) for 80°C			
Increase in length per °C			

Analysing the data

- 1 Determine the total increase in length for the full 80°C rise in temperature for each of the three metals and enter the values in the table.
- 2 Calculate the increase in length per °C rise in temperature for each of the three metals and enter these values in the table.

Note: Write these values to five decimal places.

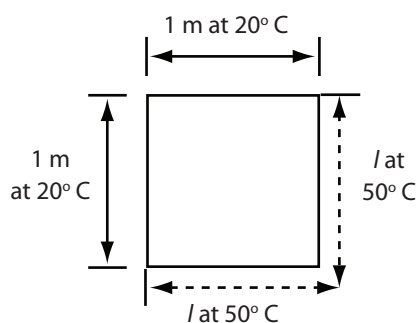
- 3 Using the new values in the last column, determine what the values for the three lengths would be at 70°C and enter these values in the appropriate column in the table.
- 4 For each °C rise in temperature, which metal appears to increase in size:
 - a the most? _____
 - b the least? _____

Interpreting the data

- 5 Using the values of increase in length per °C in the table, determine the increase in size of:
 - a a 1.0 km stretch of iron railway line that increases in temperature from an overnight low of 1°C to a maximum day temperature of 31°C.

 - b an aluminium door of height 2.0 m subjected to a temperature increase of 20°C.

- 6 A square piece of copper with sides of 1.000 00 m at 20°C had its temperature increased from 20°C to 50°C.



- a Find the new length of its sides.

- b Hence calculate the increase in area of the copper piece with the increase in temperature.

INTRODUCTION

There are two types of magnetic 'poles': north and south. Similar or like poles repel one another, unlike poles attract.

Aim

To use the results of an experiment to determine whether or not a relationship exists between the force of attraction (F) and the distance (d) between two unlike magnetic poles.

Data

Two very long, slender bar magnets were used. One was clamped in place and the other was placed in a 'cradle' that was free to swing. When the cradled magnet was brought near the clamped magnet, the attractive force caused the cradled magnet to swing towards the clamped one. It was brought back to its original position by an adjustable weight force applied by a light string over a pulley and a mass carrier. Thus, the size of the attractive force (F) was determined for a given distance of separation (d). Figure 1 shows the arrangement used.

Six different separations (d) were used and the attractive force (F) in each case was determined. The results are listed in the table below.

Note: The units are arbitrary ones as only a relationship was being investigated.

d (units)	Attractive force (F)	$\frac{1}{d}$	$\frac{1}{d^2}$	$F \times d^2$
1.0	10.00			
2.0	2.50			
3.0	1.11			
4.0	0.63			
5.0	0.40			
6.0	0.28			

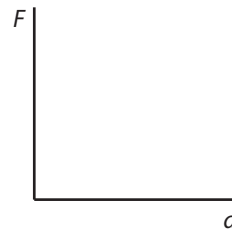
Analysing the data

- 1 From the table, what appears to happen to F as d is increased?

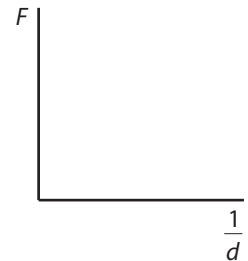
- 2 Calculate the missing values and complete the table.

- 3 Draw graphs on the axes provided to show the relationship between:

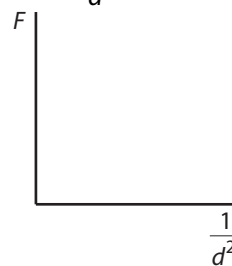
a F and d



b F and $\frac{1}{d}$



c F and $\frac{1}{d^2}$



Interpreting the data

- 4 From the shapes of the graphs, what appears to be the relationship between F and d ?

- 5 Why have very long magnets been chosen for the experiment?

- 6 What is always true about the magnetic strengths of the north and south poles of a simple bar magnet?

- 7 Suppose that two like poles (i.e. two north or two south poles) had been used for the experiment. How would the restoring force have been adjusted? Explain.

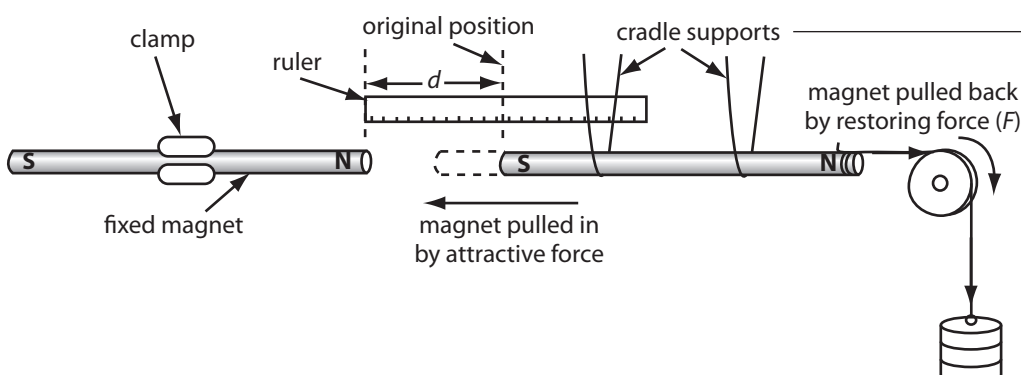


Figure 1

INTRODUCTION

The visible light section of the electromagnetic spectrum covers the range of wavelengths from approximately 400 nm to 760 nm (the red end). On either side of these limits are the ultraviolet (UV) and infrared (IR) sections of the spectrum. The UV section is the one that contains wavelengths that can be harmful to humans, particularly to the eyes.

Aim

To investigate some practical optical issues relating to the UV section of the electromagnetic spectrum (EMS). This section is divided into UV-A, UV-B and UV-C 'bands'.

Data: Set 1

The following data can be referred to for some subsequent questions. It shows the visible light section of the EMS and the neighbouring UV section. Wavelengths are given in nanometres (nm). The main graph shows the intensity (vertical) of the various wavelengths as received on the Earth's surface from the Sun. This is shown in Figure 1.

Analysing the data

1 Given that the velocity of all waves in the EMS is $3 \times 10^8 \text{ m s}^{-1}$, and that the universal wave equation is $\text{velocity} = \text{frequency} \times \text{wavelength}$, determine the values for the frequency of the waves at points A, B and C.

- A _____
B _____
C _____

2 a What is the approximate wavelength range for maximum intensity?

b What colours are included in this range?

3 Which UV band reaching the Earth is the one with most energy?

Data: Set 2

The graphs in Figure 2 show the degree of transmission of wavelengths in the range (approx.) from 290 nm to 400 nm through both plastic and glass, two materials used in the making of spectacles.

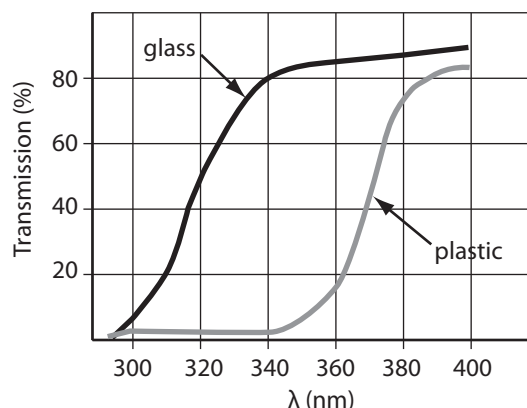


Figure 2

Analysing the data

4 Which material seems to stop almost all transmission of UV-B?

5 Which material seems to be better at blocking UV-A?

6 The graphs in Figure 2 appear to differ by very little at wavelengths of 400 nm. What does this imply about the transmission of violet light through the two materials?

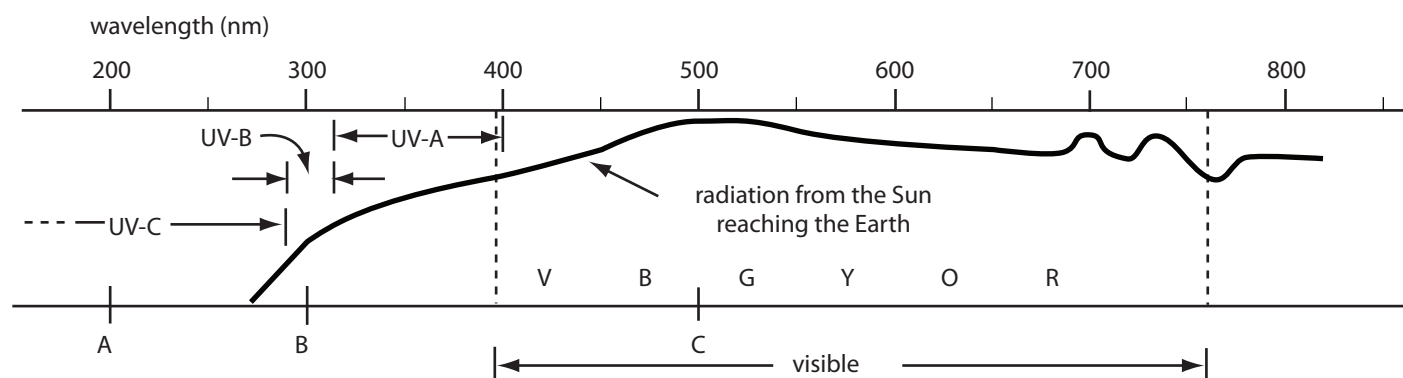


Figure 1

Data: Set 3

The graphs in Figure 3 show the amount of transmission of wavelengths for expensive and inexpensive 'UV-blocking' sunglasses.

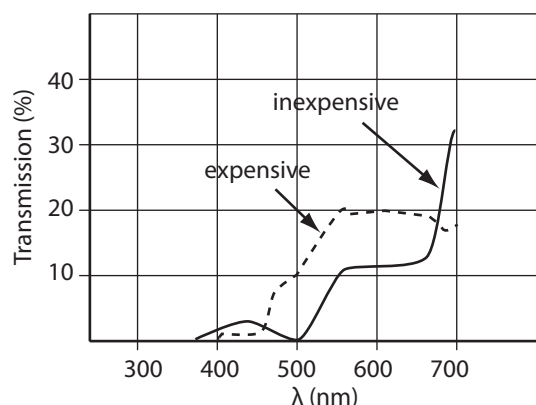


Figure 3

Analysing the data

- 7 Do the sunglasses successfully block out UV radiation? (Yes/No)

- 8 Which pair blocks out most green light?

- 9 Which pair transmits the most light of wavelength 700 nm?

Interpreting the data

- 10 A particular standards requirement for cosmetic and general purpose sunglasses regarding UV light sets out the minimum amount of radiation to be absorbed by glasses, as shown in the table.

Sunglasses	UV-A (%)	UV-B (%)
Cosmetic	20	70
General purpose	60	95

Are these standards met by:

- a the expensive sunglasses? (Yes/No)

- b the inexpensive sunglasses? (Yes/No)

Data: Set 4

The following diagrams show approximate amounts of UV radiation reflected by different surfaces. The percentages indicate the amount of the UV light that is reflected from each surface. The four surfaces are:

- a a grassy lawn
- b a paved or concreted area
- c water, e.g. a lake or ocean
- d snow and ice.

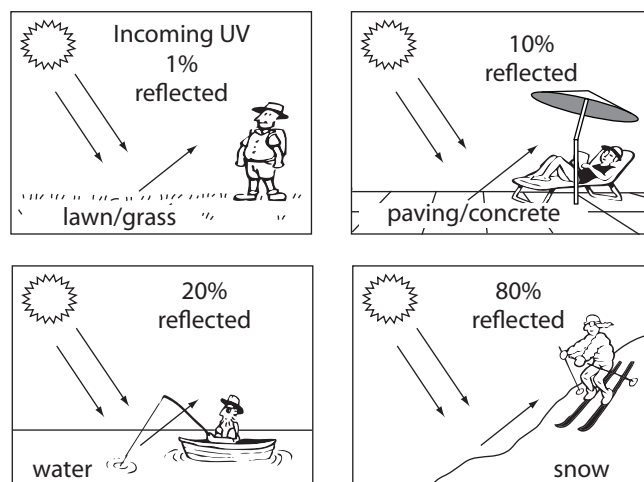


Figure 4

Interpreting the data

- 11 Professional fishermen often show a high incidence of UV-related eye problems. Suggest a reason for this.

- 12 Which environment is the one in which most exposure to UV light is experienced?

INTRODUCTION

The energy received by the Earth from the Sun and the other stars travels through space as electromagnetic radiation.

Aim

To use a few physical principles to help understand the information revealed by such radiation.

Data

When a body radiates energy it does so with wavelengths over a wide range of the electromagnetic spectrum. There is usually a narrow 'peak' in the radiation band where the intensity is at a maximum, as shown in Figure 1. The higher the temperature of the radiating body the greater the intensity of the radiation.

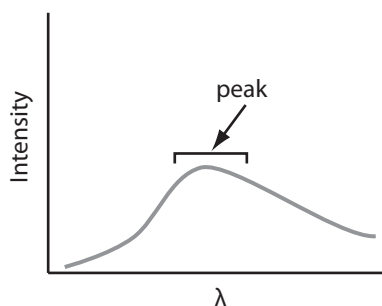


Figure 1

As the temperature increases, there is a shift in the peak towards shorter wavelengths. Figure 2 shows this with graphs for radiations at 2000 K, 4000 K, 6000 K and 8000 K. Some data of this type has been recorded in the table below.

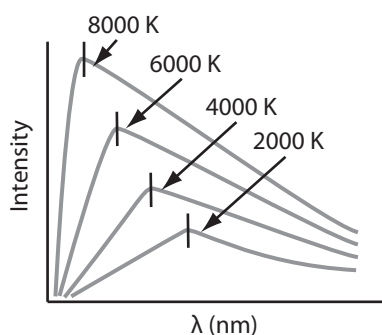


Figure 2

T (K)	λ_{peak} (nm)	$1/T$	$\lambda_{\text{peak}} \times T$
2000	1600	_____ $\times 10^{-5}$	_____ $\times 10^5$
4000	800	_____ $\times 10^{-5}$	_____ $\times 10^5$
6000	550	_____ $\times 10^{-5}$	_____ $\times 10^5$
8000	400	_____ $\times 10^{-5}$	_____ $\times 10^5$

Analysing the data

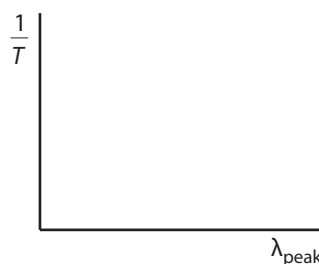
- 1 On the axes provided, draw a graph of T vs λ_{peak} .



- 2 What does the shape of the graph suggest about a relationship between T and λ_{peak} ?

- 3 Calculate the values for $1/T$ and the product $\lambda_{\text{peak}} \times T$ and complete the table. Note: As it is only the shape of the graph being analysed, the units can be ignored.

- 4 Now draw the graph of $1/T$ vs λ_{peak} . Does this graph approximate a straight line? (Yes/No) _____



- 5 What does the shape of the graph for question 4 suggest about the relationship between T and λ_{peak} ?

- 6 Is this suggested relationship in question 5 supported by the values in column 4? (Yes/No) _____

Interpreting the data

The radiation curve for 6000 K peaks in the centre of the visible area of the spectrum. For temperatures higher than this (e.g. 8000 K) the peak shifts towards the shorter wavelengths (see Figure 2).

- 7 What does this shift indicate about the proportion of:
a violet light produced at higher temperatures?

- b red light produced at higher temperatures?

- 8 What is the effect of this shift on the colour of the light emitted?

- 9 Explain how the colour of a star (e.g. blue or red) gives an indication of its temperature.

INTRODUCTION

The luminosity of a star is a measure of the total output of energy per second from the star. The energy radiates in all directions from the star and so only a tiny fraction of the energy reaches the Earth. A formula that has been derived by scientists to calculate luminosity is given below. It can be used for any star.

Aim

To use some basic data and formulae about stars to gain more information about them.

Data: Luminosity

The formula for luminosity (L) is $L \propto R^2 T^4$, where R is the radius of the star and T is its surface temperature (K). The units for L are watts (W) and normal SI units apply elsewhere. The luminosity of the Sun is approximately 3.8×10^{26} W. It is used as a reference and assigned the symbol L_0 .

Interpreting the data

The following questions should help in understanding the formula.

1 If R is doubled, by what factor does L increase?

2 If T is doubled, by what factor does L increase?

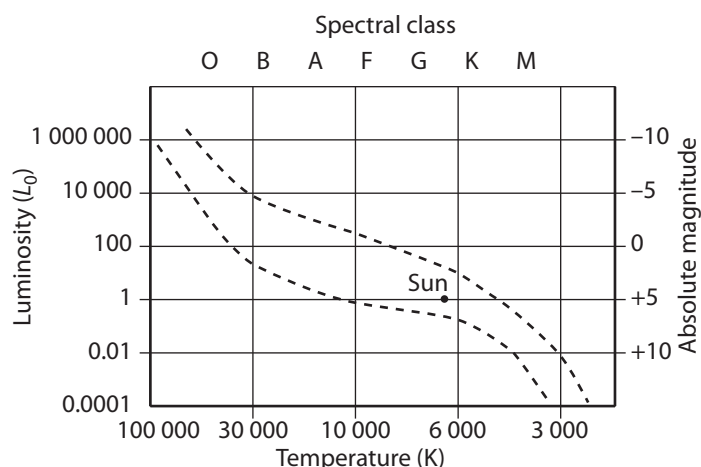
3 If both R and T are doubled, by what factor does L increase now?

4 If the star α -Centauri has $L = 8.0 \times 10^{26}$ W, it means that its luminosity is about twice that of the Sun. Does this mean that α -Centauri has both a larger radius and a higher temperature than the Sun? (Yes/No) _____

Explain your answer. _____

Data: HR diagram

A Hertzsprung–Russell diagram is one in which the luminosity (or a closely related quantity called ‘absolute magnitude’) is plotted against the temperature (or ‘spectral class’) of stars. It forms a sort of graph with many stars represented.



Hertzsprung–Russell diagram

Analysing the data

- 5 What appears to be the equivalent ‘absolute magnitude’ for a luminosity of L_0 ?

- 6 If a star has a surface temperature of 16 000 K, what would be its spectral class?

- 7 The star Betelgeuse is classed as an M2 type star with an extremely high luminosity of the order of 10^4 .
 - a What does this mean about the amount of energy it radiates compared with the Sun?

 - b How is the answer to part a explained, considering that the Sun is a much hotter star?

- 8 Plot the position of these stars on the HR diagram and decide whether or not they are ‘main sequence’ stars.
 - a Spica: absolute magnitude = -2.4
spectral class B _____
 - b Antares: absolute magnitude = -5.1
spectral class M1 _____
- 9 What would be the approximate temperatures of the two stars in question 8?
 - a Spica _____
 - b Antares _____

INTRODUCTION

Johannes Kepler (1571–1630) and Isaac Newton (1642–1727) are two scientists who made great contributions in many areas of science. Newton was born 12 years after Kepler's death, so the two did not work together. The second part of this exercise shows how Newton's later laws can be used to arrive at Kepler's third law from a completely different point of view, thus substantiating Kepler's work.

Aim

To look at some of the laws proposed by Kepler and Newton and their interrelationships.

Data: Kepler's laws

Kepler's study of planetary motion around the Sun enabled him to enunciate three laws:

Law 1. *The planets orbit the Sun in elliptical orbits with the Sun at one focus.* (See Figure 1.)

Law 2. *The line joining the planet and the Sun (radius) sweeps out equal areas in equal periods of time.* (See Figure 2.)

Law 3. *The square of the orbital period of a planet is directly proportional to the cube of its average distance from the Sun, R , i.e. $\frac{T^2}{R^3} = k$.*

Because of the large numbers involved, the unit of distance used for R is the astronomical unit AU, where 1 AU = distance from Sun to Earth. The unit for period T is the time it takes the Earth to orbit the Sun (= Earth years).

elliptical orbit

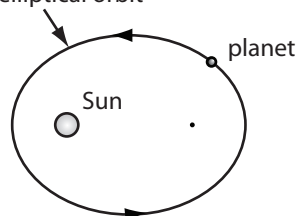


Figure 1

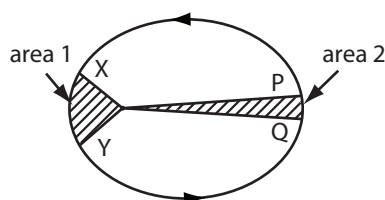


Figure 2

Analysing the data

- As the two foci in Figure 1 get closer together, what happens to the elliptical shape?
- In Figure 2 the arc XY of the planet's orbit is longer than its arc at PQ. What feature of the planet's motion changes to make this happen?
- Since T and R both have values of 1 for the Earth, what would be the value of k for the Earth in Kepler's third law?

This value is true for any body orbiting the Sun.

Data: Newton's laws

Two of Newton's great contributions to physics were (1) his law of universal gravitation and (2) the laws of motion. The first of these laws is expressed as follows:

$$F_a = \frac{Gm_1m_2}{d^2}$$

where m_1 and m_2 are the two masses in question and d is their distance of separation; G is the constant of proportionality for the equation. The force, F_a , is always an attractive one. This situation is shown in Figure 3. The relevant law of motion here is the second law, which is expressed as $F = ma$ for an object undergoing an acceleration.

A third equation used here is the one expressed as $a = \frac{v^2}{r}$ for an object undergoing uniform circular motion where v is the magnitude of the orbital velocity and r is the radius of its orbit. The force to keep an object in circular motion is the centripetal force, which is always directed towards the centre of the circle. This situation is shown in Figure 4.

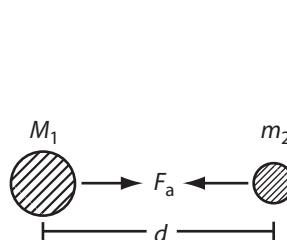


Figure 3

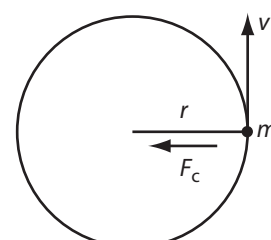


Figure 4

Analysing the data

- Express the force of gravitational attraction of a planet (mass m_p) to the Sun (mass M_s) where the planet orbits at a mean distance of R .
- Express the centripetal force required to keep a planet in circular motion around the Sun at a distance of R , using the orbital velocity value v .
- Since the gravitational attractive force is the centripetal force, the two expressions from questions 4 and 5 can be equated. Do this and so eliminate F from the equation. Reduce the resulting equation to its simplest form.
- In time T the planet completes one revolution around the Sun and has a constant speed of v . What is the distance moved in terms of R ?

- 8 Express this speed v (i.e. magnitude of velocity) as a distance per time relationship. It should be in terms of R and T .
- 9 Now substitute the value of v from question 8 into the equation derived in question 6. This rather involved equation can then be rearranged to make $\frac{T^2}{R^3}$ the subject. Do this and observe what terms are now present on the right-hand side.
- 10 If question 9 has been done correctly then it can be seen that all terms on the right-hand side of the equation are constants. List these constants without listing their values.

The combination of constants in questions 9 and 10 themselves make up a constant. This leads to the formula proposed by Kepler long before Newton:

$$\frac{T^2}{R^3} = k$$

Data: Kepler’s third law

Some information regarding the orbiting of the Sun by three planets is listed in the table below. The measurements of period and distance are modern, accurate measurements, but they are in ‘Earth days’ and millions of kilometres respectively rather than the units used in Kepler’s third law.

Planet	Venus	Mars	Saturn
Period in Earth days	224.70	686.98	10 759.20
Period in Earth years			
Distance from Sun (millions of km)	108	228	1 426
Distance in AU (= R)			
$\frac{T^2}{R^3}$			

The idea here is to ‘check’ that Kepler’s third law is valid for these planets. To do this (1) the unit for period has to be converted from Earth days to T (the value for T being 365.26 days) and (2) the unit of distance has to be converted from millions of km to AU (1 AU = 149×10^6 km).

Analysing the data

- 11 Convert the period and distance values to the appropriate units and enter the values into the respective columns in the table.
- 12 Now substitute the new values into the formula in the last column and calculate $\frac{T^2}{R^3}$. Kepler’s third law claims that these values are approximately constant. Does this appear to be true? (Yes/No)
- 13 The value of k for the Earth is 1.000. Are the values calculated in question 12 compatible with the Earth’s value? (Yes/No)
- Should they be? (Yes/No)
- Explain.
- 14 Another heavenly body that is thought to be a satellite of the Sun appears to have a period, in Earth days, of 4332.59 and is at an average distance from the Sun of 778×10^6 km. Determine whether or not it is a satellite of the Sun.

Interpreting the data

- 15 If the value of k was to be determined for the orbit of the Moon around the Earth, would its value be expected to be the same as the values calculated previously? (Yes/No)
- Explain.
- 16 If a communications satellite were to be put into orbit around the Earth would its value of k be similar to that of the Moon? (Yes/No).
- Explain.

INTRODUCTION

In attempts to reduce carbon dioxide emissions from motor vehicles, manufacturers have promoted the electric car. The idea is to produce a car that is powered by electricity from rechargeable batteries and so produces no carbon dioxide emissions when running. A potential car is one that has a bank of batteries that can be charged overnight.

Aim

To investigate and compare the emissions from a petrol-powered car and those from a similar electric car powered by electricity from a coal-burning power station.

Data

A particular small electric car has the following claims. Its batteries can be charged in 8 hours from a normal household 240 V supply using a 15 A output. Once the batteries are charged, the car can travel a distance of about 160 km if no electrical accessories (e.g. air conditioner, radio, lights) are on. Some additional data will be supplied as needed.

Analysing the data

- I Using the following formula:

$$\text{Power} = \text{Potential difference} \times \text{Current}$$

determine the power input to the electric car during charging.

- 2 Calculate the total energy input to the car in the charging period.

- 3 The energy calculated in question 2 is the energy consumed by the car as it travels 160 km. Calculate the energy consumed per 100 km.

Some more data

As coal (which is treated here as carbon) is burned to produce energy to make electricity, there is some energy loss at each of the various stages between the burning and the final use of electricity in the home. For this exercise it is assumed that there is a 20% overall loss of energy. If 12.0 g of carbon is burned, 44.0 g of carbon dioxide (CO₂) and 393 kJ of energy (if there is no loss) are produced.

- 4 Calculate the amount of usable energy from burning 12.0 g of carbon, allowing for the 20% loss.
- 5 Calculate the mass of carbon that has to be burned to allow the electric car to travel 100 km.
- 6 Since 12.0 g of carbon produces 44 g of CO₂, how much CO₂ must be produced for the car to travel 100 km?
- 7 Now calculate the mass of CO₂ produced for the car to travel 1.0 km.

Even more data

A small petrol-driven car uses 5.0 L per 100 km. The density of petrol can be taken to be 0.7 g m L⁻¹. The energy released from the petrol as it burns can be assumed to be about 5000 kJ for each 100 g of fuel burned. For every 1.0 g petrol burned approximately 3.1 g of CO₂ is produced.

- 8 Calculate the mass of petrol burned by the car every 100 km.
- 9 Calculate the mass of CO₂ produced by the car every 100 km.
- 10 Determine the mass of CO₂ produced per kilometre as the car travels.

Interpreting the data

- 11 Which vehicle, electric or petrol, appears to be the less polluting with regard to CO₂ emissions considering this data?
- 12 Before it can be said that the electric car is definitely less polluting than the petrol car, what other considerations need to be addressed in order that a fair comparison can be made?

INTRODUCTION

The ever-increasing concerns about the rapidly rising levels of the greenhouse gas carbon dioxide has caused great interest in producing electrical energy by means other than fossil fuels. Such energy production is intended to create no such gas as there is no combustion involved. There are many forms of alternative energy in resources such as the wind, the Sun, the tides and the energy stored in 'hot rocks' under the Earth's surface.

Aim

To investigate the implementation of a few of these alternative energy resources in some Australian situations.

Preliminary data

A few basic ideas need to be consolidated here:

- a** The basic unit of power is the watt, defined as $1 \text{ W} = 1 \text{ J s}^{-1}$. The joule is a small unit of energy, so the watt is a small unit of power.
- b** A more practical unit of electrical energy for household purposes is the kilowatt-hour, which is defined as the amount of energy produced (or consumed) when a device of power 1 kW operates for a time of 1 h. This is the commonly used unit for household consumption.

Analysing the preliminary data

- 1** Determine the number of joules in a kilowatt-hour.

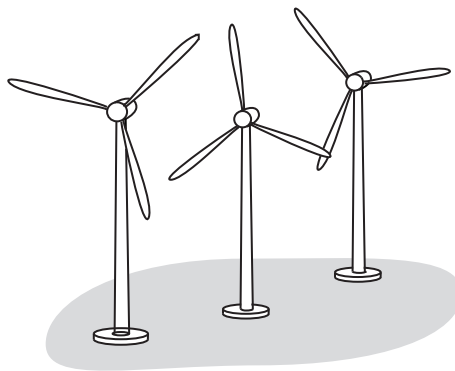
- 2** An example of the energy consumption of a household is 7000 kW h in a year. Supposing that 1 kW h costs 18c. What would be the cost of electrical energy per year for the household?

- 3** A 'greenhouse gas factor' on an electricity account is '1.068', which means that the carbon dioxide produced by a coal-burning power station is 1.068 kg for every 1.0 kW h of energy produced. What mass of carbon dioxide would be produced by the household in question 2 in 1 year?

- 4** The volume of 44.0 g of carbon dioxide gas is approximately 25 L. What volume of carbon dioxide would be produced by the mass from question 3?

Data: Wind power

One alternative way of producing electricity is by using wind-powered turbines. Groups of such turbines make up 'wind farms'. Australia has about 30 such farms (capable of producing in excess of 50 MW) either operating, being constructed or with planning approval.



Analysing the data

- 5** For the household that uses 7000 kW h energy in a year, determine the rate of energy usage in kW.

- 6** How many such households could be supplied energy by a 50 MW wind farm?

- 7** A huge wind farm near Broken Hill in NSW will have the capacity, when fully operational, to supply 440 000 homes with electricity from its 598 turbines. How many homes will one turbine supply?

- 8** If the energy consumption rate for each home is as calculated in question 5, determine the power output of each turbine.

- 9** Calculate the total power output of the wind farm.

- 10** The calculations for the electrical supply above assume a constancy of energy output. Suggest a reason why this may not always be the case.

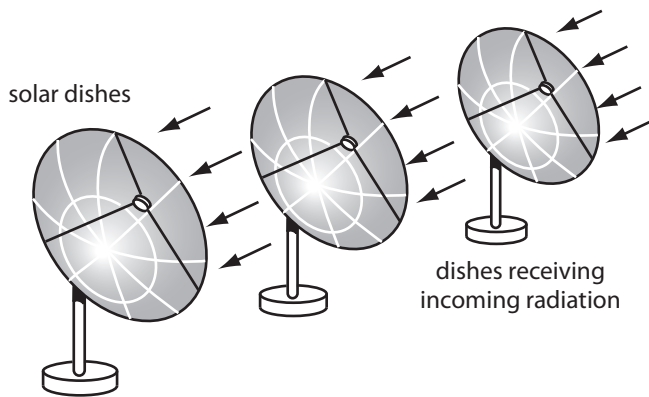
- 11** A typical power output of a traditional coal-burning power station is usually much higher than that of a wind farm such as the one near Broken Hill. Such an output might be 1500 MW. How many wind farms would be needed to equal this?

Data: Solar power

The small outback town of Windorah in Queensland has set an example in electricity production by solar power. The small town of 100 residents has 60 buildings (homes and businesses) powered by electricity from 5 dishes. On sunny days, the power output of the 5 dishes is enough to fulfil the total daytime requirements of the 100 residents. Over a year the 5 dishes will produce 300 000 kW h of electricity.

Analysing the data

- 12** If each of the 60 buildings consumes, say, 7000 kW h of electricity per year what is the yearly requirement of the town?



- 13** Will the dishes alone be able to accommodate these requirements? (Yes/No) _____
- 14** What is the yearly output of one dish? _____
- 15** What is the average power in kilowatts of one dish? _____
- 16** In question 5 the average power requirement of a single similar building was calculated. Using this figure, determine how many homes one dish could power fully. _____
- 17** On the basis of the above, calculate the number of homes (including businesses) that the 5 dishes could fully power in a year. _____

- 18** The claim is that the output of the 5 dishes will fulfil the daytime requirements of all of the residents, but the answer to question 17 means that this does not occur over a year. Explain why this is the case.
- _____
- _____

- 19** Even though the solar energy does not fulfil the town's total requirements over a year, it is anticipated that there will be a saving of some 10 000 L of diesel fuel over that time. Diesel fuel has normally been the source of energy for the town's electricity supply. This means that 300 tonnes of carbon dioxide will be saved from going into the atmosphere in that time. Using the value in question 4 that 44.0 g of carbon dioxide has a volume of approximately 25 L, calculate the volume of carbon dioxide going into the atmosphere each year that is saved by the use of solar energy.
- _____
- _____

Interpreting the data

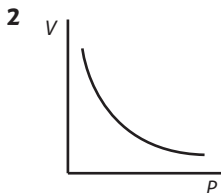
- 20** When comparing the number of homes supported by one solar dish at Windorah with the number supported by one wind-powered turbine near Broken Hill, it is apparent that the wind-powered turbine can provide for many more households than the solar dish. From the figures above, determine how many times more 'effective' one wind turbine is than one solar dish.
- _____
- 21** The photovoltaic cells that directly produce electricity usually have a very low voltage output. Many such cells may need to be connected together to produce sufficient voltage to be useful for domestic purposes. There are two systems of connection for such cells: in series and in parallel. Which system would have to be used to make the voltages of the cells add together?
- _____
- 22** In many solar energy 'farms' the Sun's energy is not used to produce electricity directly, as with photovoltaic cells, but it is concentrated to a point to create very high temperatures, sometimes in the thousands of degrees. Suggest a way by which this concentration of energy might produce electricity in an alternative way.
- _____
- _____
- _____

Answers

1 The importance of graphs

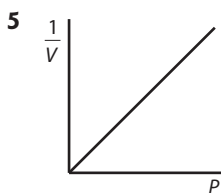
1 a $y \propto x$ or $y = kx$

b $y \propto \frac{1}{x}$ or $y = \frac{k}{x}$ or $yx = k$



3 No

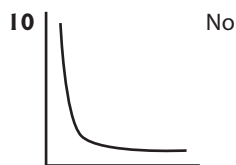
4 $\frac{1}{V} = 0.05, 0.10, 0.15, 0.20, 0.25, 0.30$



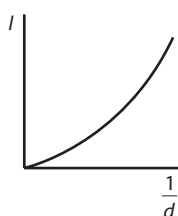
6 Yes 7 $\frac{1}{V} \propto P$ or $PV = k$

8 $PV = \text{constant}$

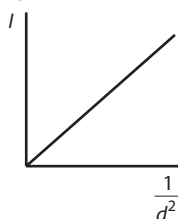
9 $PV = 40.0, 40.0, 40.2, 40.0, 40.0, 39.6$; Yes



11 a $\frac{1}{d} = 1.00, 0.50, 0.33, 0.25, 0.20$



b $\frac{1}{d^2} = 1.00, 0.25, 0.11, 0.06, 0.04$



12 Yes; I vs $\frac{1}{d^2}$ 13 $I \propto \frac{1}{d^2}$ or $I = \frac{k}{d^2}$

2 Uncertainty in measurement

1 B: Yes; C: No

2 a 1.65 cm b 1.75 cm

3 1.7 ± 0.05 cm 4 3.30 cm

5 3.50 cm 6 3.4 ± 0.1 cm

7 1.40 cm 8 1.60 cm

9 1.5 ± 0.1 cm

10 a Half smallest division (least count)

b Are added c Are added

3 Swimming underwater

1 a F_1 : 43.3 N up, 25.0 N left

b F_2 : 43.3 N down, 25.0 N left

2 a 0 N b -50 N

3 Swimmer pushes on water, reaction force of water on swimmer is to the right.

4 There is an equal and opposite force exerted by the water.

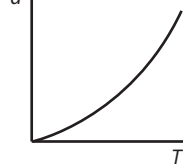
5 700 N 6 700 N 7 50 N

8 50 N 9 It increases it.

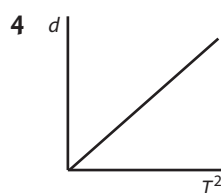
10 Mass (hence weight), of water displaced is greater and so upthrust is greater.

4 Motion down an inclined plane

1 d 2 No



3 $T^2 = 0.25, 1.00, 2.25, 4.00, 6.25, 9.00$



5 Yes 6 $s \propto t^2$ or $s = kt^2$

7 0.5 8 Half the acceleration

9 1.0 m s^{-2}

10 a 2.6 m b 2.8 s

11 Displacement

12 Yes. Since a , u and t are known, v can be calculated.

13 Its mass

14 Distance would increase

5 The chiming clock

1 31.36 J 2 25.09 J

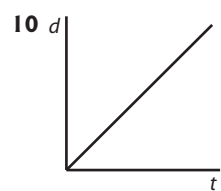
3 Sound energy of chimes, friction of moving parts.

4 $5.2 \times 10^{-5} \text{ J s}^{-1}$ 5 Power, watts

6 0.566 m 7 $1.6 \times 10^{-4} \text{ m s}^{-1}$

8 No. Speed is constant, velocity changes with direction.

9 95.1 m



11 Zero

12 Every half hour

6 The coefficient of friction

Weight	F_r	$F_f:F_r$
10	10	0.50
15	15	0.51
20	20	0.51
25	25	0.50
35	30	0.50
30	35	0.50

2 Constant 3 $\mu = F_f/F_r$

4 a 25 N b 3.2 kg

5 3.25 kg

6 Once wheels lock friction becomes less and braking less effective.

7 Very low

7 Determining the acceleration due to gravity I

1 One quarter 2 0.39 s

3 9.86 m s^{-2} 4 Just a little larger

5 a Cord slows fall

b Friction in pulleys/pivot

6 a Increase b Increase

7 No atmosphere decreases time of swing.

8 0.97 s

8 Determining the acceleration due to gravity II

1 $M_2 a_g$ down (LHS), T up (RHS)

2 $M_1 a_g - T = M_1 a$; $T = M_1 a_g - M_1 a$

3 $T - M_2 a_g = M_2 a$; $T = M_2 a_g + M_2 a$

4 $a_g = \frac{a(M_1 + M_2)}{M_1 - M_2}$

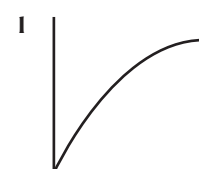
5 0.234 m s^{-2} 6 9.6 m s^{-2}

7 a Smaller b Friction in pulley

8 a Decrease b Increase

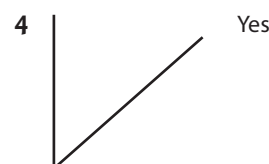
c Unchanged

9 The swinging of a simple pendulum



2 a No b No

3 T^2 : 1.61, 2.43, 3.24, 4.04, 4.84, 5.66, 6.45, 7.29, 8.06, 8.88



5 $T^2 \propto l$ or $T^2 = k \times l$

6 $T \propto \sqrt{l}$ or $T = k\sqrt{l}$ 7 Gravity

8 Tension in string or rod and gravitational force

9 Zero

10 a A or B b C

10 Handicap races

1 a 9.90 m s^{-1} b 10.15 m s^{-1}

2 0.5 m 3 21.8 m

4 a 9.95 m s^{-1} b 9.70 m s^{-1}

5 3, 1, 2, 4 6 13.89 m s^{-1}

7 24 min 8 1465 s and 1513 s

9 13.65 m s^{-1}

11 The conservation of energy

1 $a = F/m$ 2 $W = mas$

3 $KE = mas$ 4 $KE = \frac{1}{2}mv^2$

5 a $W = 80 \text{ J}$, $a = 10 \text{ m s}^{-2}$,
 $v = \sqrt{80} \text{ m s}^{-1}$, $KE = 80 \text{ J}$

b Yes

6 a mgh b 0 c mgh

7 Gravity 8 mgh

9 $\sqrt{2}gh$

10 mgh 11 mgh

12 $PE = mgh - mgx$ $v_C = \sqrt{2}gx$
 $KE = mgx$ Total energy = mgh

13 Same, mgh 14 Agree

12 Some physics and aircraft

1 0.62 m s^{-2} 2 233.34 kN

3 $3.78 \times 10^6 \text{ N}$

4 a Lessen it
b Greater speed or wingflap angle

5 $9.45 \times 10^7 \text{ kg m s}^{-1}$ 6 0.926 m s^{-2}

7 902.85 N 8 35.96 s

9 541 710 J 10 15.06 kW

11 Much less 12 Small one

13 A lot of fuel has been consumed.

13 Torque

1 No confusion with 'millinewtons'

2 Fig. 1: 10.0 Nm, Fig. 2: 5.0 Nm,
Fig. 3: 3.42 Nm

3 No

4 a i 500 N ii 750 Nm
iii 400 N iv 800 Nm

b Girl's

5 Towards the girl 6 To the right

7 75 Nm each 150 Nm 8 $Bll \times \frac{b}{2}$ each

9 $Bll \times b$ 10 Same B , l and l

11 $Bll \times \text{Area of loop}$

12 a 1.20 N b $1.8 \times 10^{-2} \text{ Nm}$

c $3.6 \times 10^{-2} \text{ Nm}$

13 $7.2 \times 10^{-2} \text{ Nm}$

14 The work of Henry Cavendish

1 $4.7 \times 10^{-8} \text{ N}$

2 Very sensitive and accurate considering tiny value of force measured.

3 $a_g = \frac{GM_E}{(R_E)^2}$ 4 9.83 m s^{-2}

5 Value is extremely close

6 N kg^{-1}

7 Density of interior rocks much greater than 5.4 kg m^{-3}

8 a Prevention of drafts from moving small spheres.

b Enable readings of the very small movements to be measured.

15 Circular motion

1 Constant change in direction.

2 $v_a + \Delta v = v_b$

3 $\Delta v = v_b + (-v_a)$ 4 4.8 m s^{-1}

5 Towards centre of circle.

6 Towards centre of circle.

7 76 m s^{-2} 8 7.6 N

9 Centripetal 10 Tension in string

11 a increase b increase

12 Lessen it 13 Path Y

14 No longer any force in horizontal plane

15 First law

16 The work of James Prescott Joule

1 2058 J 2 82 320 J

3 20 400 cal 4 $4.04 \text{ J} = 1 \text{ cal}$

5 A little higher

6 a Heat loss to surroundings

b Water gains some M.E., etc.

7 To prevent adding more energy as masses were lifted.

17 Pressure

1 a 250 cm^2 b 200 cm^2
c 80 cm^2

2 a 2.5×10^{-2} b 2.0×10^{-2}
c 8×10^{-3}

3 a 7.84×10^2 b 9.80×10^2
c 24.50×10^2

4 19.6×10^6

5 Approx. 100× greater.

6  7 No

8 Approx. 4× 9 Explode

18 Pressure in a liquid

1 10^4 kg 2 10^4 Pa

3 $11 \times 10^4 \text{ Pa}$ 4 10 m

5 16A (water) + 1A (air) = 17A

6 0.2 L 7 1.33 m s^{-1}

8 It would increase it.

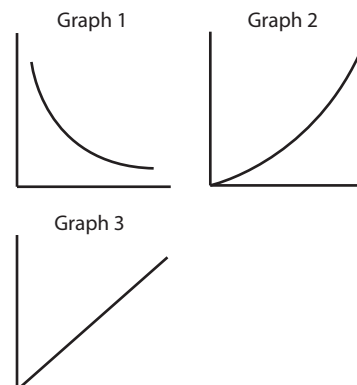
9 Dissolved salts increase density.

10 11.76 m

19 The force between electric charges

1 $\frac{1}{d}$: 0.050, 0.033, 0.025, 0.020, 0.017, 0.014

$\frac{1}{d^2}$ ($\times 10^{-4}$): 25.0, 11.1, 6.3, 4.0, 2.7, 2.0



2 Graph 3

3 $F \propto \frac{1}{d^2}$ or $F = \frac{k}{d^2}$

4 Both small mass and good insulators

5 No. Size of forces the same, but attractive.

20 Determining the value of an unknown resistance

1 A, C, B 2 A, D, B

3 Same

4 a $PD = I_1 R_1$ b $PD = I_2 R_3$

5 $I_1 R_1 = I_2 R_3$

6 a $PD = I_1 R_2$ b $PD = I_2 R_4$

7 $I_1 R_2 = I_2 R_4$ 8 $\frac{R_1}{R_2} = \frac{R_3}{R_4}$

9 $R_1 = \frac{R_2 R_3}{R_4}$ 10 No

11 3 ohms 12 3:4

13 a No b 3:4 c Same

14 Ratio of one to the other.

21 Transformers

- 1 LCD TV – 69.6, 67.5
Laptop – 360, 65.0
Power station – 2.00×10^6 , 1.98×10^6
Light – 144, 128
- 2 97%, 18%, 99%, 89%
- 3 Very high voltages involved
- 4 **a** 3 **b** 1, 2, 4
- 5 Conversion to heat
- 6 Low voltage is safer to use.
- 7 **a** 6.48×10^5 J **b** 5.31×10^5 J

22 Electricity and heating

- 1 85.7°C **2** 576 kJ
- 3 4.2 **4** 5.1 hours
- 5 Yes
- 6 **a** 100.8 kJ **b** 17.5 A

23 Wave transmission I

- 1 Decreases
- 2 **a** Rubber **b** Diamond
- 3 Rubber
- 4 11.67, 3.18, 2.24, 3.02, 3.14, 3.07, 0.36, 0.03
- 5 **a** No **b** No
- 6 Water has no resistance to shear, like all liquids.
- 7 They are not mechanical waves and do not need a medium to pass through.

24 Wave transmission II

- 1 Fig. 2: Refractions towards the normals.
Fig. 3: Refractions away from the normals.
- 2 Modulus of compression. Velocity is increasing with depth, but so is density.
- 3 Decrease
- 4 Much denser
- 5 Modulus of shear has zero value.
- 6 Modulus of shear and modulus of compression increase with depth in mantle, while density increases. Mantle–core boundary suggests very dense core, possibly liquid.

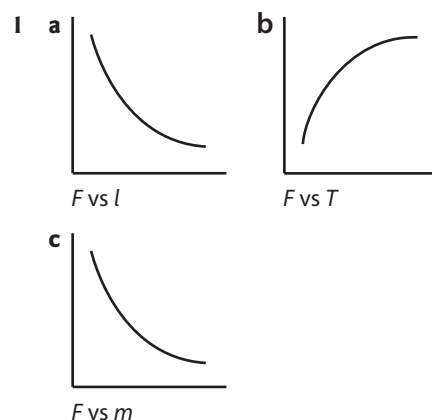
25 Total internal reflection and optical fibres

- 1 RI: Water = 1.33, Plastic 2 = 1.52, Plastic 3 = 1.49
 θ_c : Flexible glass = 43.23° , Plastic 1 = 43.98°
- 2 **a** No **b** Yes **c** No **d** Yes
- 3 Many combinations possible, e.g. water around any other substance.
- 4 **a** Flexible glass and plastic 1 (outside)

b Plastic 2 and water (outside)

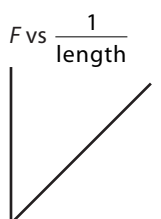
- 5 A single frequency rather than multiple frequencies, refracted differently.
- 6 Electrical

26 Vibrating strings



- 2 **a** No **b** No
- 3 Experiment 1

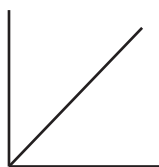
$\frac{1}{l}$	0.01	0.02	0.04	0.08
$F \times l (\times 10^4)$	2.56	2.56	2.56	2.56



Experiment 2

$\sqrt{T}$	5.0	10.0	20.0	25.0
$\frac{F}{\sqrt{T}}$	20	20	20	20

F vs $\sqrt{T}$



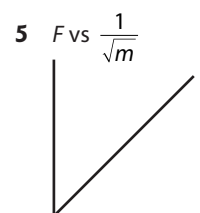
Experiment 3

$\sqrt{m}$	0.1	0.2	0.3	0.4
$\frac{1}{\sqrt{m}}$	10.0	5.0	3.3	2.5
$F \times \sqrt{m}$	10.0	10.0	9.9	10

F vs $\sqrt{m}$

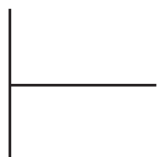


- 4 **a** Yes **b** First two



- 6 Yes **7** Yes
- 8 **a** $F \propto \frac{1}{l}$ **b** $F \propto \sqrt{T}$
- c** $F \propto \frac{1}{\sqrt{m}}$
- 9 Yes **10** Make pitch higher.

27 Water temperature change

- 1 Water under very high pressure.
- 2 Huge difference in scale.
- 3 **a** 0.95 **b** 0.5°C , 7.0°C
- 4 As the temperature increases, density decreases.
- 5 Volume increases, but mass is unchanged.
- 6 Mass vs temp.  Volume vs temp. 
- 7 Graph C: Mass does not change. Graph D: Volume increases as water expands.
- 8 Density drops as water volume increases, so outward pressure on bottle.

28 The expansion of metals

- 1 $\Delta L = 0.000\,96$, $0.002\,08$, $0.001\,36$
- 2 ΔL per $^\circ\text{C} = 0.000\,012$, $0.000\,026$, $0.000\,017$
- 3 $1.000\,60$, $1.001\,30$, $1.000\,85$
- 4 **a** Al **b** Fe
- 5 **a** 36 cm **b** 0.104 cm
- 6 $L = 1.000\,51$ m, $A = 0.001\,02$ m²

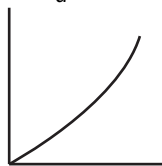
29 Forces between magnetic poles

- 1 It decreases.
 - 2
- | | | | | | | |
|-----------------|-------|-------|------|-------|-------|-------|
| $\frac{1}{d}$ | 1.00 | 0.50 | 0.33 | 0.25 | 0.20 | 0.17 |
| $\frac{1}{d^2}$ | 1.00 | 0.25 | 0.11 | 0.06 | 0.04 | 0.03 |
| $F \times d^2$ | 10.00 | 10.00 | 9.99 | 10.08 | 10.00 | 10.08 |

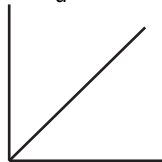
3 a F vs d



b F vs $\frac{1}{d}$



c F vs $\frac{1}{d^2}$



4 $F \propto \frac{1}{d^2}$

5 To minimise effect of other ends.

6 They are equal.

7 Force would be repulsive, so restoring force would be in opposite direction.

30 Aspects of the electromagnetic spectrum

1 $A = 1.5 \times 10^{15}$
 $B = 1.0 \times 10^{15}$
 $C = 6.0 \times 10^{14}$

2 a Approx. 450–600 nm

b Blue, green, yellow

3 UV-A 4 Plastic

5 Plastic 6 About the same.

7 Yes 8 Inexpensive

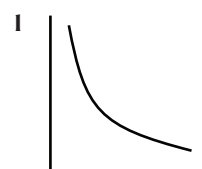
9 Inexpensive

10 a Yes b Yes

11 The high reflectivity of water would subject fishermen to long-term high doses of UV light.

12 Snow and ice

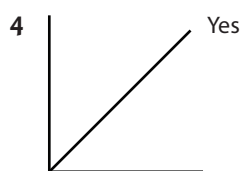
31 Radiant energy from stars



2 Inverse

3 $\frac{1}{T} = 50.0, 25.0, 16.7, 12.5$

$\lambda_{\text{peak}} \times T = 32, 32, 33, 32$



5 Definitely inverse, $\lambda_{\text{peak}} \times T = k$

6 Yes

7 a Increases b Decreases

8 More bluish

9 Blue stars = higher temperatures;
 red stars = lower temperatures

32 Luminosity of stars and the HR diagram

1 $\times 4$ 2 $\times 16$

3 $\times 64$

4 No, could be either factor or both.

5 +5 6 A/B

7 a About 10^4 times that of the Sun

b Radius of Betelgeuse is huge compared to Sun.

8 a Yes b No

9 a 20 000 K b 2500 K

33 The laws of Kepler and Newton

1 Approaches circular shape.

2 Its speed along XY increases.

3 $k = 1.00$ 4 $F = \frac{Gm_p M_s}{R^2}$

5 $F = \frac{m_p v^2}{R}$ 6 $v^2 = \frac{GM_s}{R}$

7 $d = 2\pi R$ 8 $v = \frac{2\pi R}{T}$

9 $\text{RHS} = \frac{4\pi^2}{GM_s}$

10 Constants are 4, π , G , M_s .

11 Periods = 0.615, 1.881, 29.456
 Distances = 0.725, 1.530, 9.570

12 Ratios = 0.992, 0.989, 0.990. Yes.

13 Yes. Yes. All orbit the Sun, so k should be 1.0.

14 $\frac{T^2}{R^3} = 0.989$, so it is.

15 No. No, it is a different system, so different k .

16 Yes. Both orbit the Earth, so same k .

34 The electric car

1 3600 watts

2 $10\,368 \times 10^4$ joules

3 64 800 kJ 4 314.4 kJ

5 2.473 kg 6 9068 g

7 90.7 g 8 3500 g

9 10 850 g 10 108.5 g

11 Electric

12 Energy for maintenance, replacement parts (e.g. batteries) etc.

35 Aspects of alternative energy

1 3.6×10^6 J 2 \$1260

3 7.476×10^3 kg 4 4.25×10^6 L

5 0.8 kW h^{-1} 6 62 500

7 735.8 8 588.6 kW

9 352 MW

10 Periods of little or no wind, breakdowns, etc.

11 4.3 12 4.2×10^5 kW h

13 No 14 60 000 kW h

15 6.85 kW 16 8.56

17 42.8

18 No output from dishes in times of no light, e.g. at night, on cloudy days.

19 1.7×10^8 L 20 86

21 In series

22 High temperature boils water. Steam drives turbine.

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Interpreting data in Senior Physics

This senior secondary Physics resource includes self-contained units of work that will refine students' ability to read, describe, analyse, manipulate and interpret data. Students create diagrams, complete tables and form thoughtful written responses about topics such as alternative energy, determining acceleration due to gravity, torque, electricity and heating, and the electric car. The theories of scientists Newton, Kepler, Cavendish and Joule are also linked to the data and unit scenarios. Each unit follows a stepped process that will ensure that students develop appropriate science method skills. Extended answers have been included to assist with marking and to promote a complete understanding of the topic.

Topics include

The swinging of a simple pendulum
The conservation of energy
Physics and aircraft
Torque
The work of Henry Cavendish
Circular motion
The work of James Prescott Joule

Pressure in a liquid
The force between electric charges
Transformers
Electricity and heating
Wave transmission
Total internal reflection and optical fibres

Associated titles in the Instant Lessons series

- Interpreting data in Senior Biology
- Interpreting data in Senior Chemistry
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